

UNIVERSIDAD TECNICA
FEDERICO SANTA MARIA

Modeling and simulation of flexible systems: A tutorial

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Outline

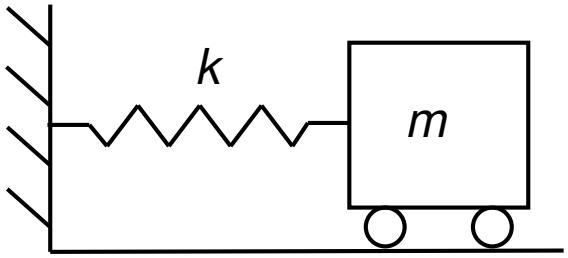
1. Introduction

2. Modeling of linear flexible systems

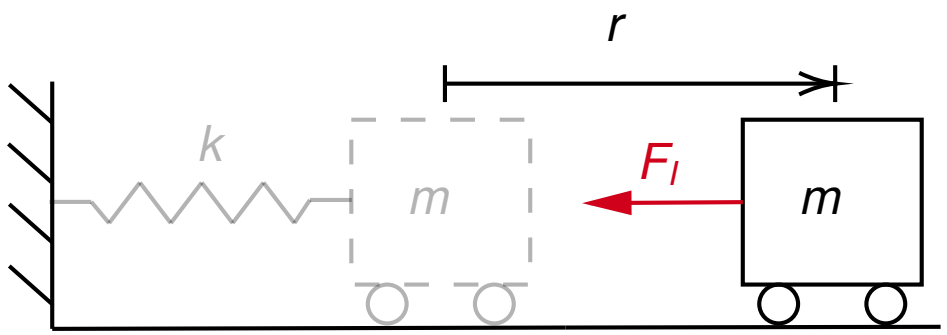
3. Finite element discretizations

1. Introduction

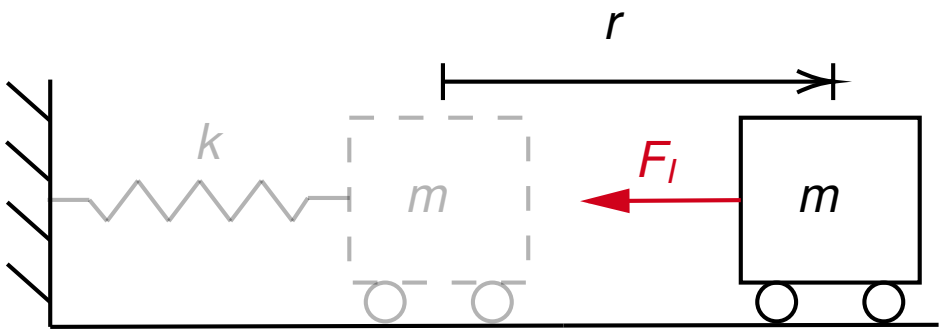
Newton's method



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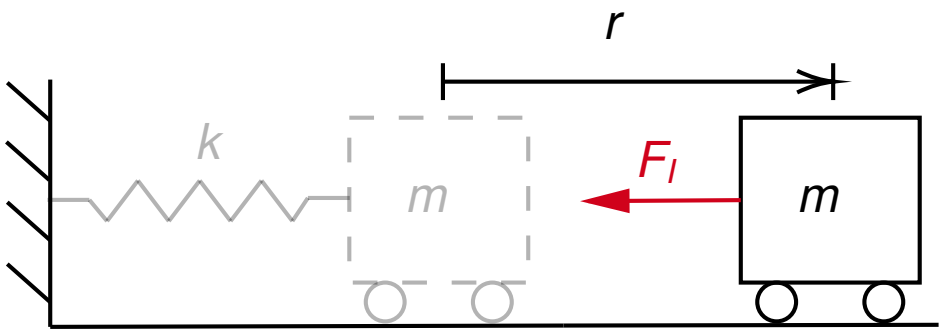


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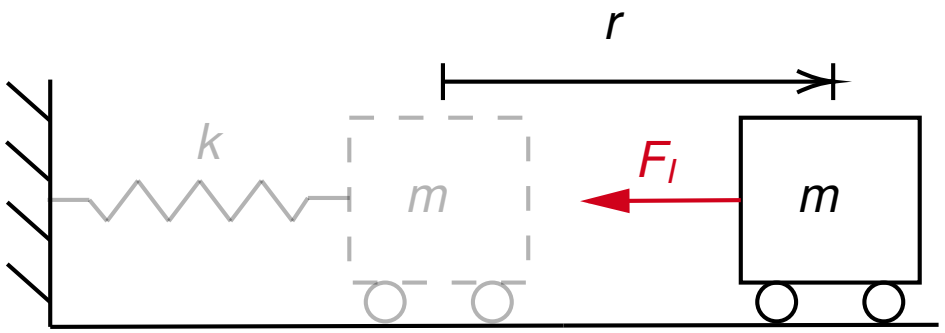
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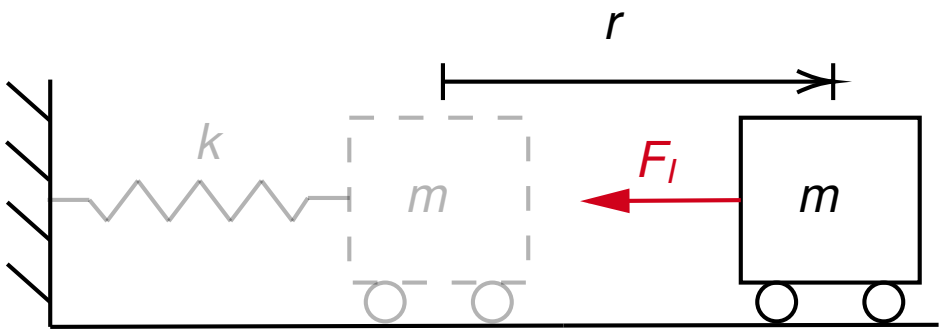
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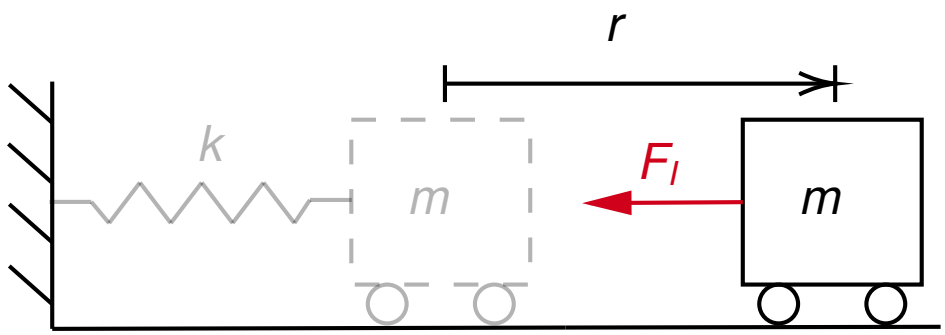
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$$\sum F = m\ddot{r} \quad \Longrightarrow \quad -F_l = m\ddot{r}$$

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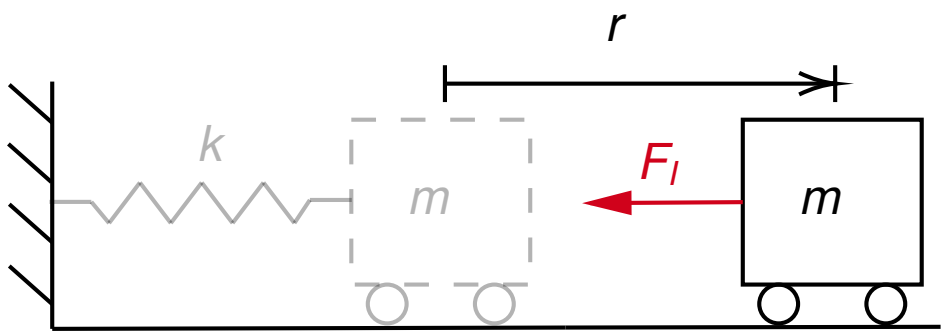
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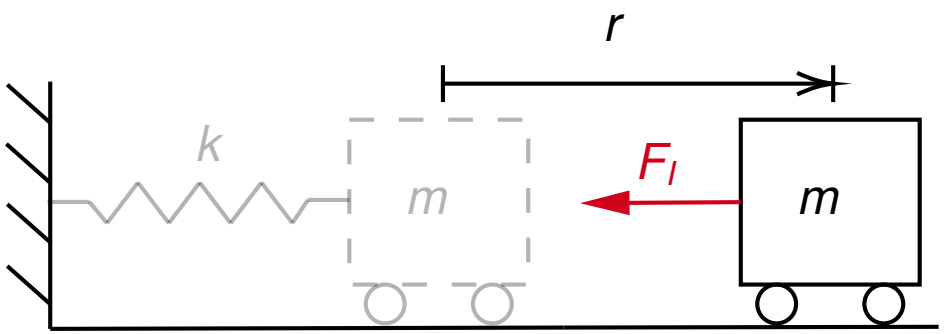
- Using a constitutive law for the internal force, i.e., $F_I = k\,r$ (Hooke's law), we obtain the final model:

$$m\ddot{r} + k\,r = 0$$

Euler-Lagrange equation

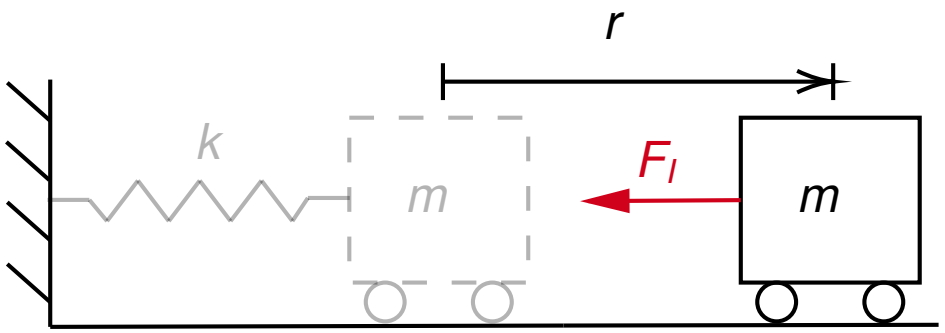


Euler-Lagrange equation



- We define the Lagrangian functional as: $L(r, \dot{r}) = T(\dot{r}) - U(r)$, where $T(\dot{r})$ is the kinetic energy and $U(r)$ is the elastic energy.

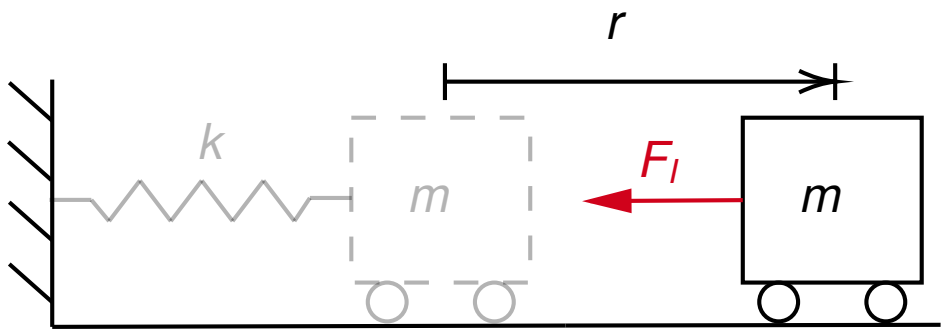
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- The model is obtained by substituting $L(r, \dot{r})$ into the Euler-Lagrange equation:

$$\frac{\partial L}{\partial r} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}}\right) = 0 \tag{1}$$

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- For the system above: $T(\dot{r}) = \frac{1}{2}m\dot{r}^2$ and $U(r) = \frac{1}{2}kr^2$ (Hooke's law). Then:

$$(-k\,r) - \frac{d}{dt}(m\dot{r}) = 0 \quad \Longleftrightarrow \quad m\ddot{r} + k\,r = 0$$

Hamilton's principle

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- To derive a model, it is necessary to solve the stationary condition of $\mathcal{S}$, i.e., $\delta\mathcal{S} = 0$, where $\delta(\cdot)$ is the variational operator.

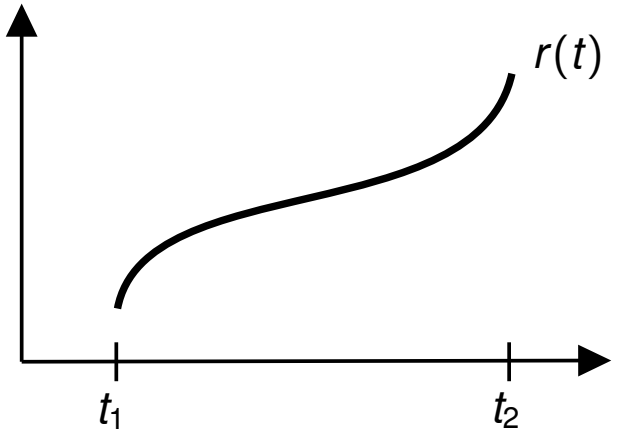
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- When the action functional includes the effect of external forces (through the virtual external work δW_E), the principle is often referred to as Hamilton's principle, and is usually stated as:

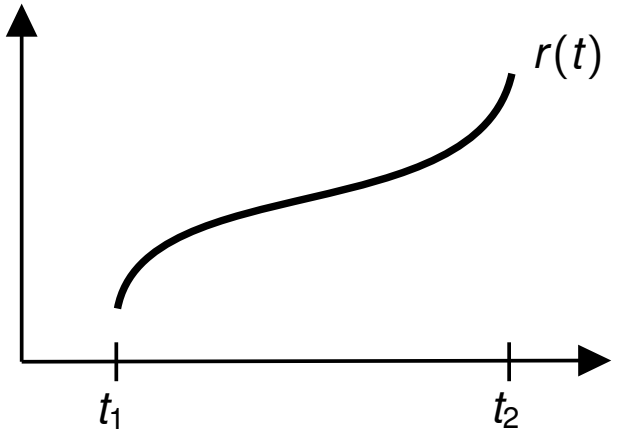
$$\int_{t_1}^{t_2} [\delta(T - U) + \delta W_E]\, dt = 0, \tag{2}$$

$$\delta r(t_1) = \delta r(t_2) = 0. \tag{3}$$

Refresher: The variational operator $\delta(\cdot)$



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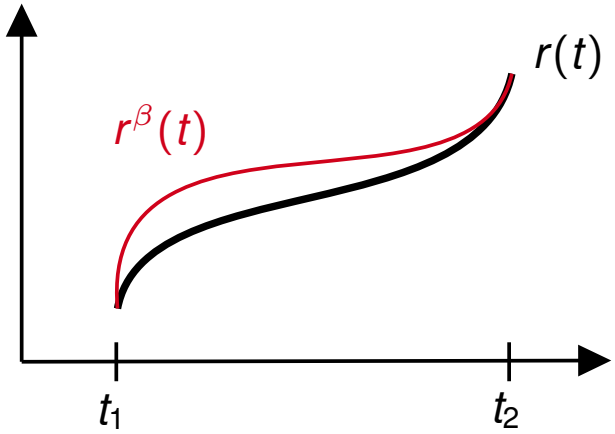


- A small perturbation of a function $r(t)$ in the interval $[t_1\ t_2]$ is denoted as:

$$r^{\beta}(t) = r(t) + \beta\eta(t)$$

with $\eta(t)$ an arbitrary function satisfying $\eta(t_1) = \eta(t_2) = 0$ and $\beta \in \mathbb{R} \to 0$.

Refresher: The variational operator $\delta(\cdot)$

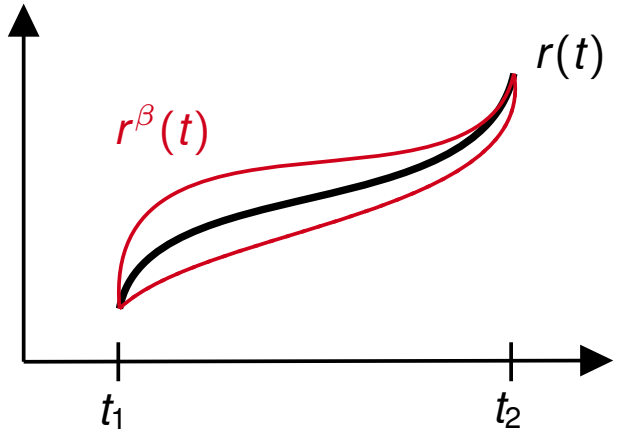


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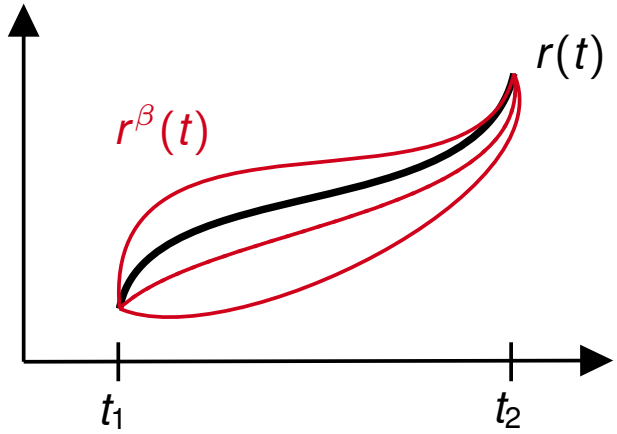


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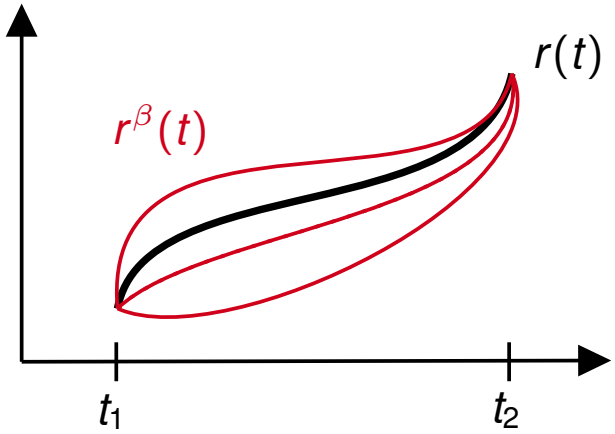


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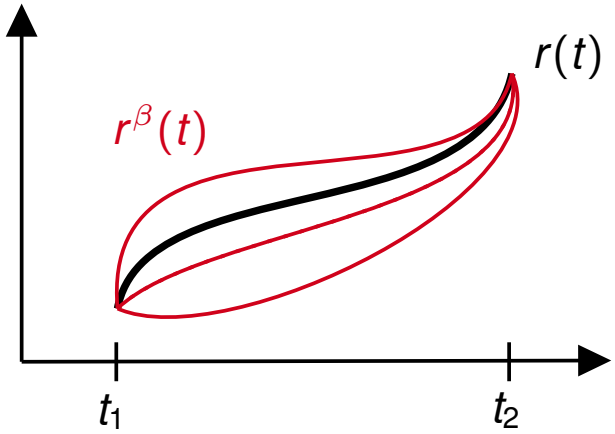
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- A small perturbation of a functional $\mathcal{S}(r, \dot{r})$ in the interval $[t_1 \;\; t_2]$ is denoted as: $\mathcal{S}^\beta = \mathcal{S}(r^\beta, \dot{r}^\beta)$.

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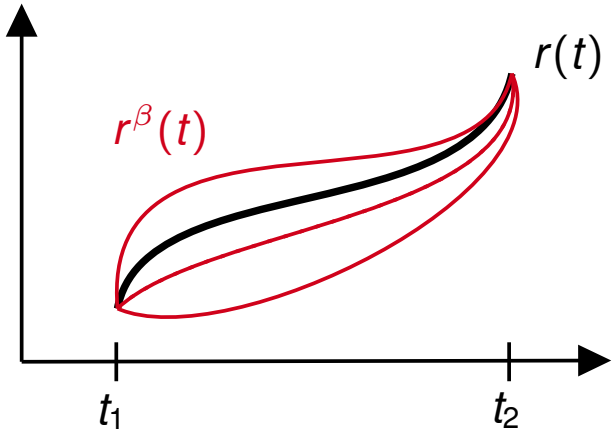
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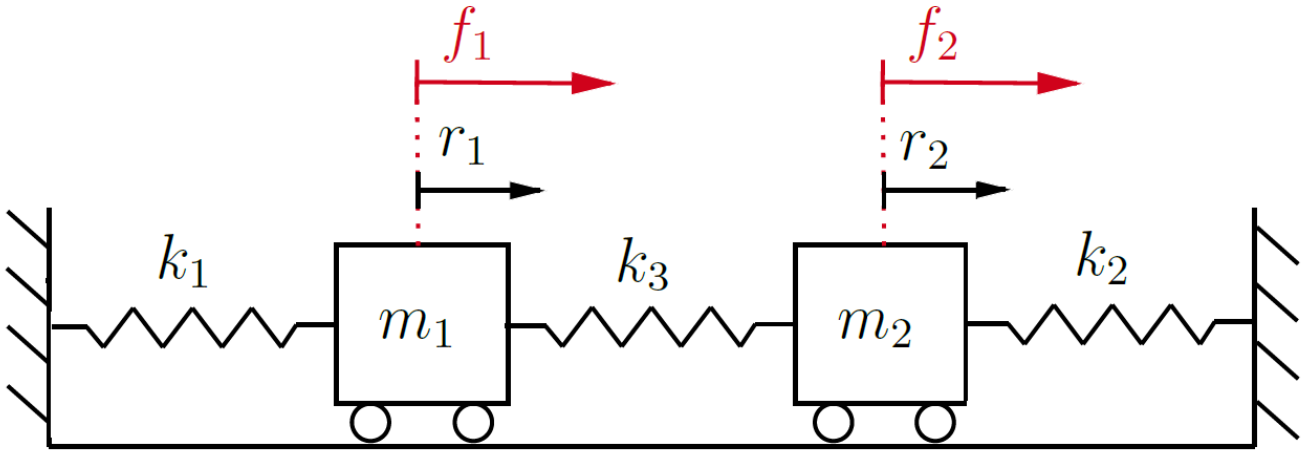
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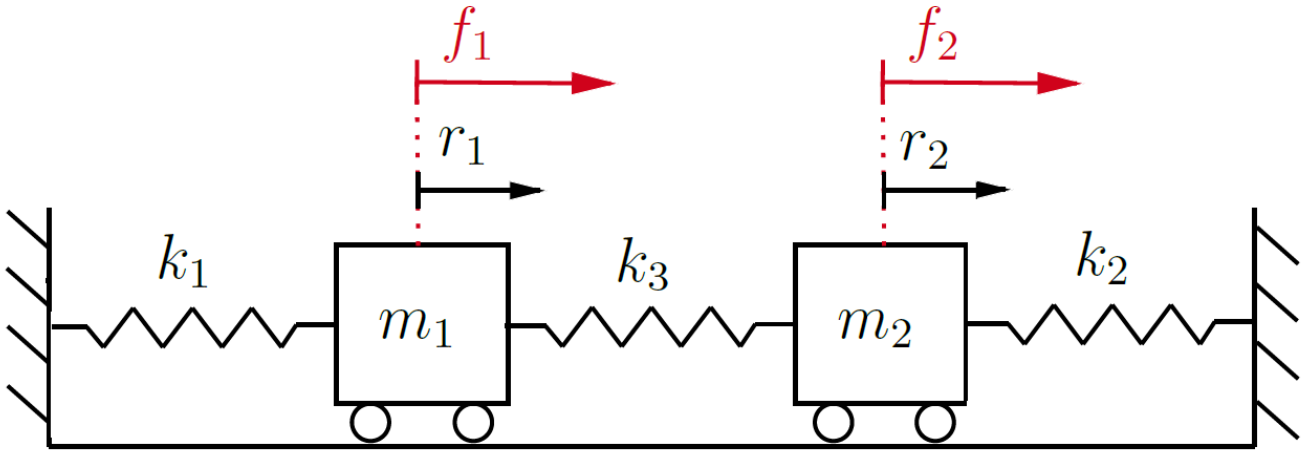
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- **Important:** Notice that $\delta r(t) = \eta(t)$, then $\delta r(t)$ is an arbitrary function that satisfies $\delta r(t_1) = \delta r(t_2) = 0$.

Lumped parameter system

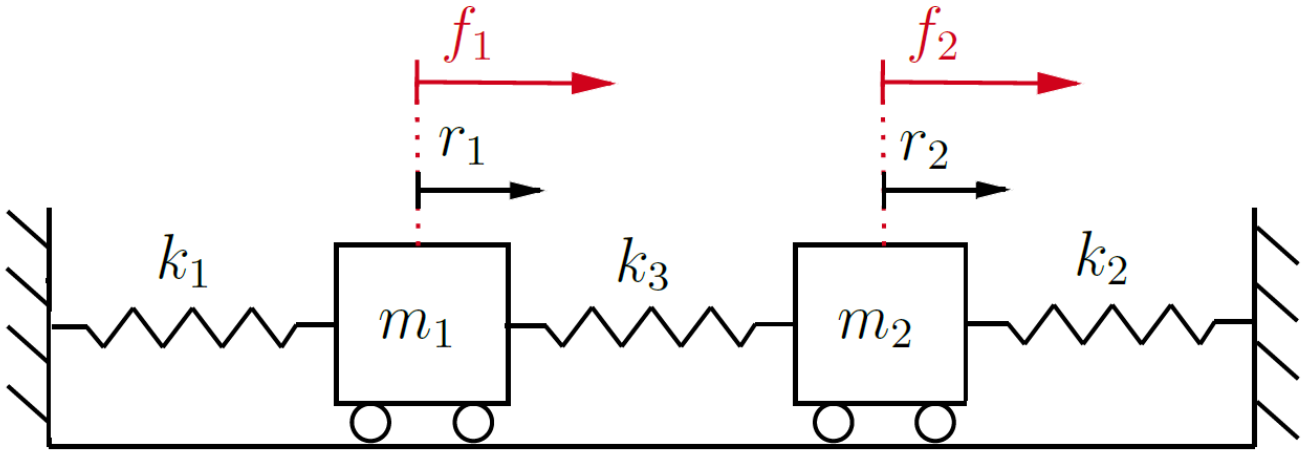


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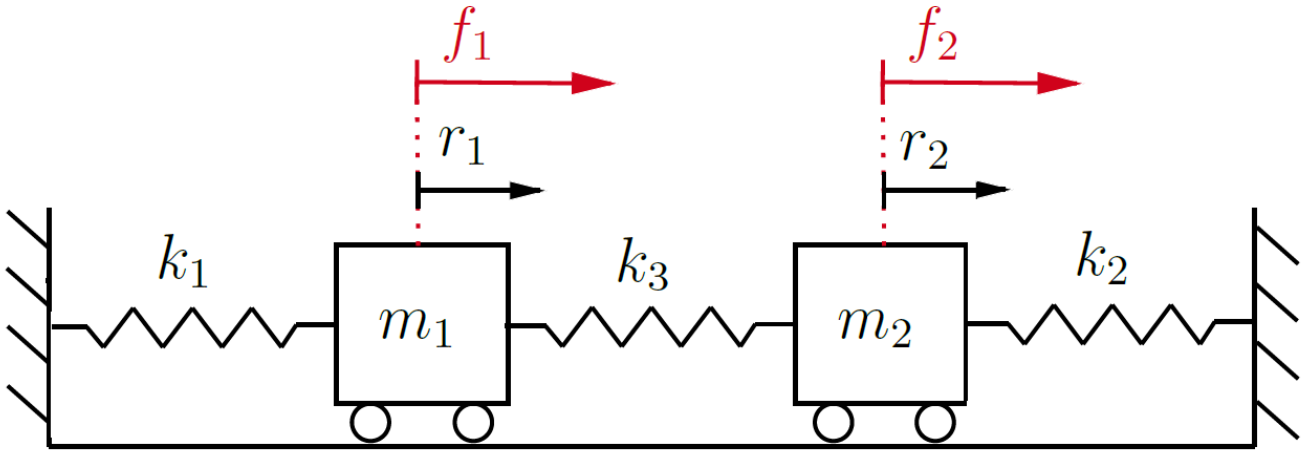
$$T(\dot{r}) = \frac{1}{2} m_1 \dot{r}_1^2 + \frac{1}{2} m_2 \dot{r}_2^2$$

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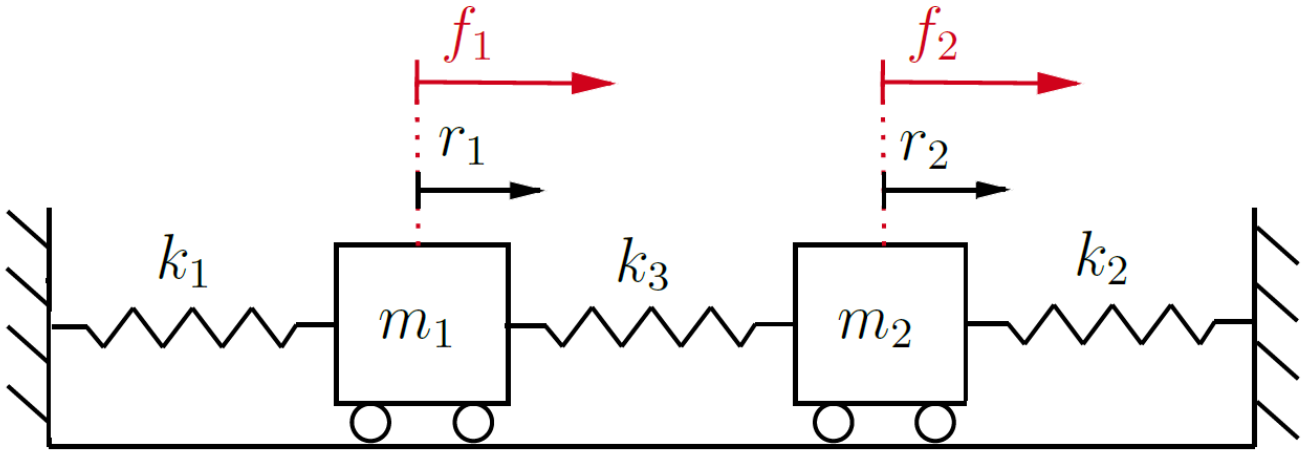
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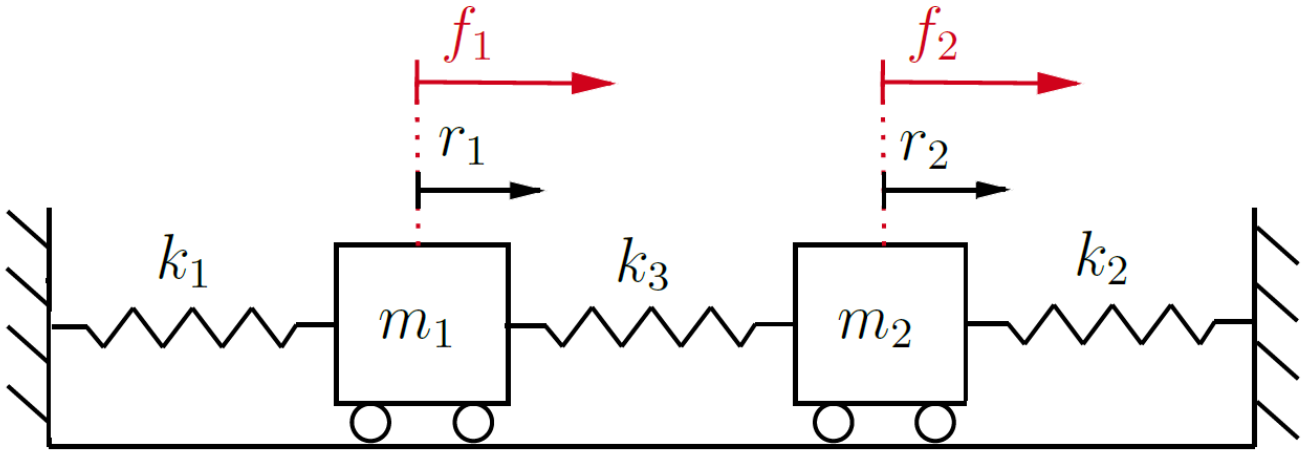
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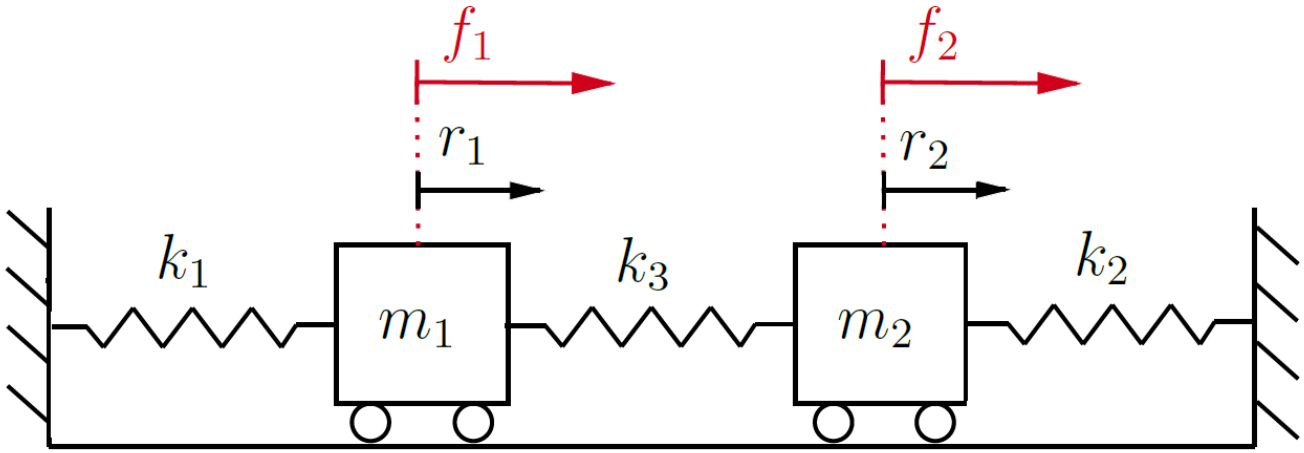


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$$\int_{t_1}^{t_2} [\delta(T - U) + \delta W_E]\, dt = 0, \tag{4}$$

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$$\rightarrow \int_{t_1}^{t_2} [\delta \dot{r}^\top M \dot{r} - \delta r^\top K r + \delta r^\top F]\, dt = 0$$

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$$\rightarrow \int_{t_1}^{t_2} [\underbrace{\delta \dot{r}^\top M \dot{r}}_{\text{i.b.p.}} - \delta r^\top K r + \delta r^\top F]\, dt = 0$$

$$\rightarrow \int_{t_1}^{t_2} \delta r^\top [-M\ddot{r} - Kr + F]\, dt + (\delta r^\top M \dot{r})\Big|_{t_1}^{t_2} = 0$$

Hamilton's principle

$$\int_{t_1}^{t_2} [\delta(T - U) + \delta W_E]\, dt = 0, \tag{4}$$

$$\delta r(t_1) = \delta r(t_2) = 0, \tag{5}$$

For the previous system we have:

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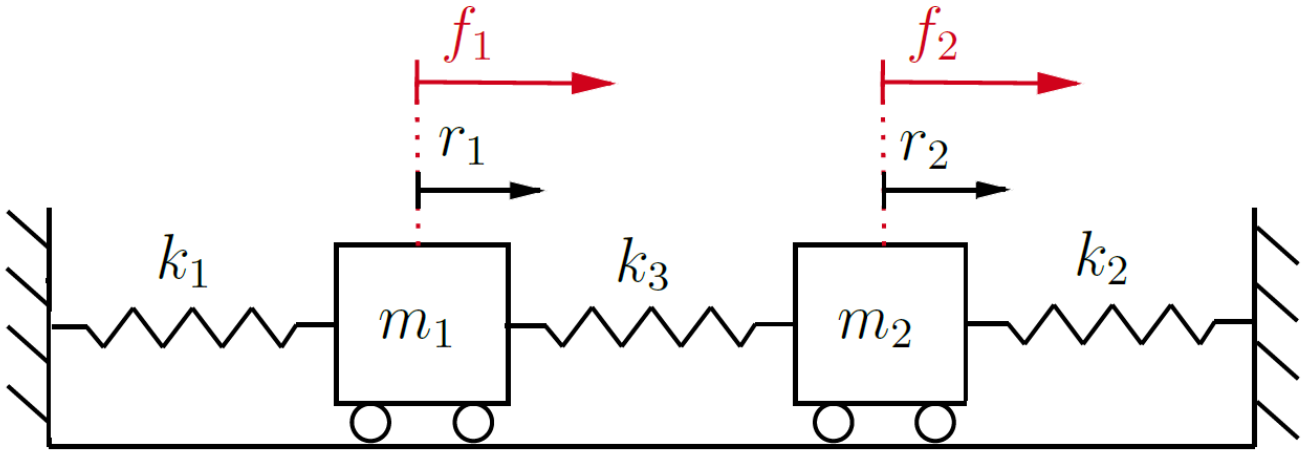
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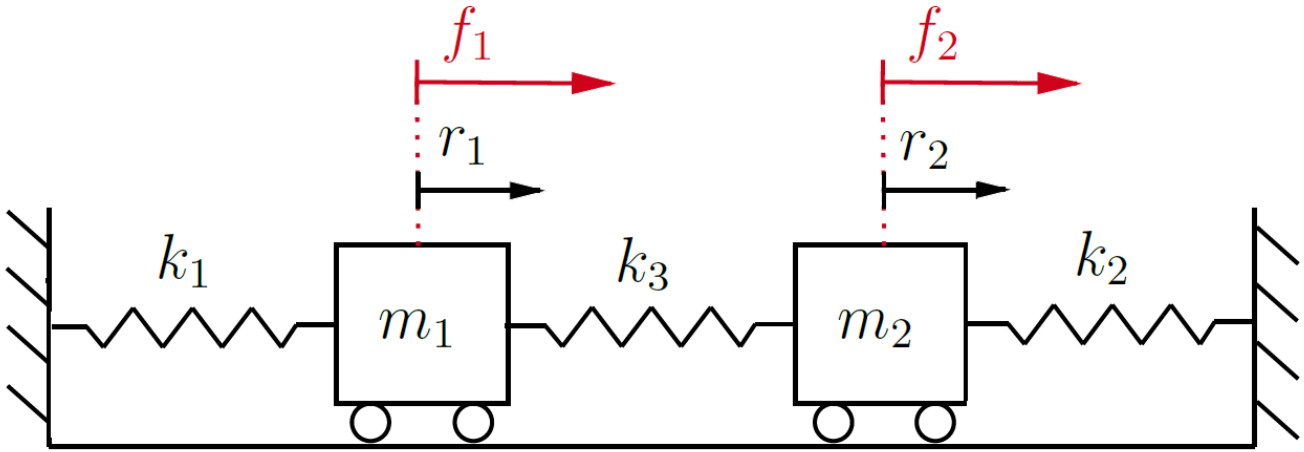
$$M\ddot{r} + Kr = F \tag{6}$$

Lumped parameter system: PHS representation



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Lumped parameter system: PHS representation



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Defining $p = M\dot{r}$, model (6) can be rewritten as:

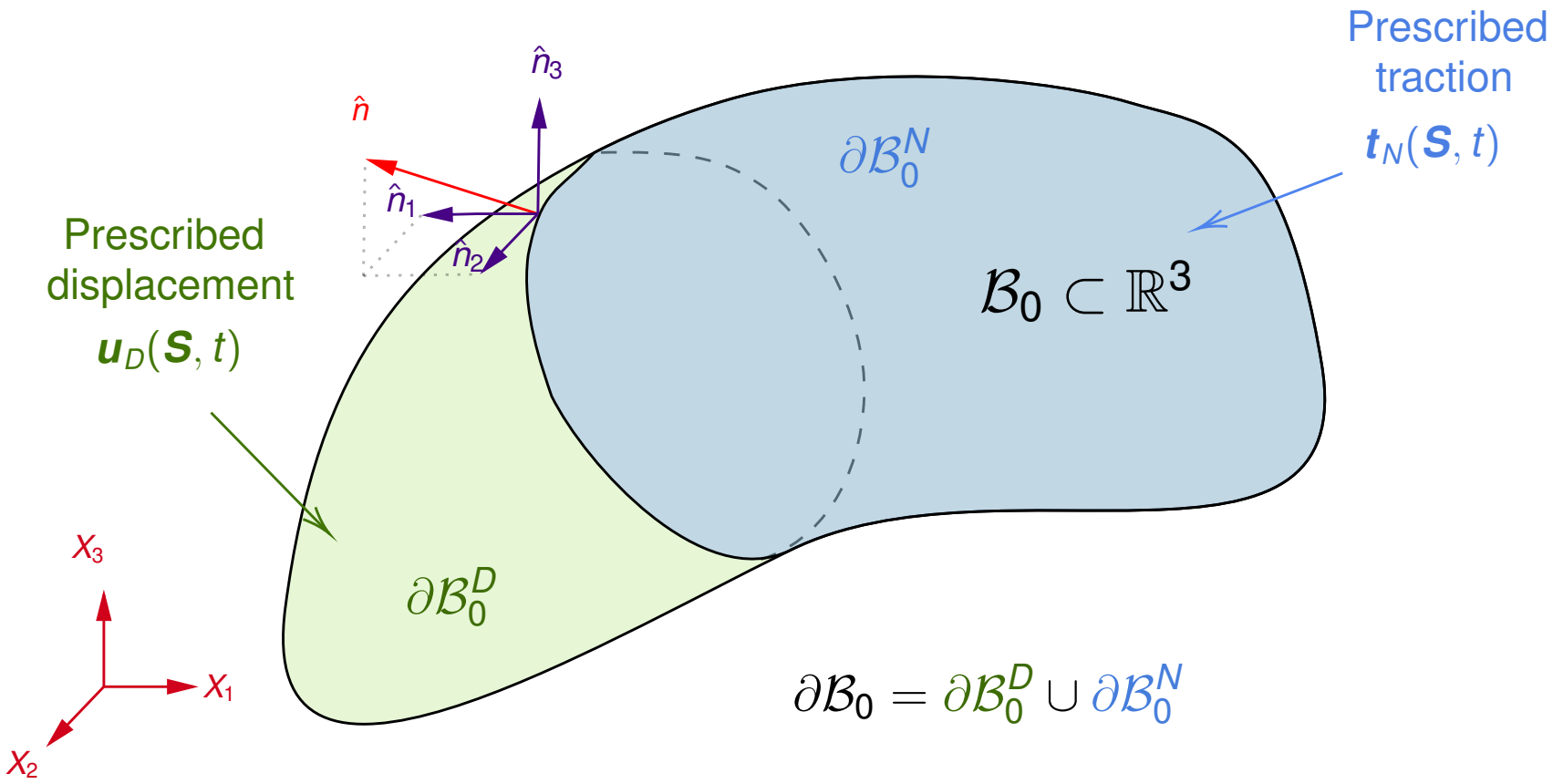
$$\underbrace{\begin{bmatrix}\dot{p}\\ \dot{r}\end{bmatrix}}_{\dot{x}} = \underbrace{\begin{bmatrix}0 & -I\\ I & 0\end{bmatrix}}_{J=-J^\top}\underbrace{\begin{bmatrix}M^{-1}p\\ Kr\end{bmatrix}}_{\nabla H(x)} + \underbrace{\begin{bmatrix}I\\ 0\end{bmatrix}}_{G} F$$

$$y = G^\top \nabla H(x)$$

$$H(x) = \tfrac{1}{2} p^\top M^{-1} p + \tfrac{1}{2} r^\top K r$$

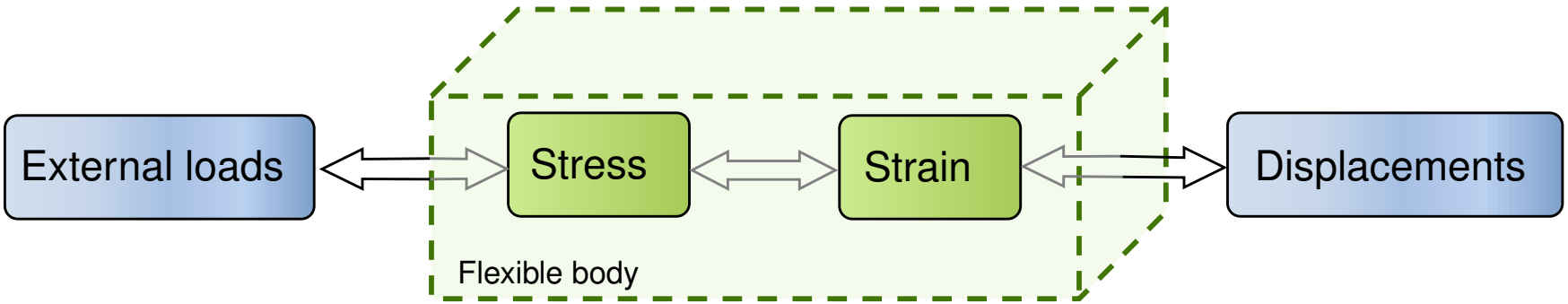
2. Modeling of linear flexible systems

Background: Flexible bodies

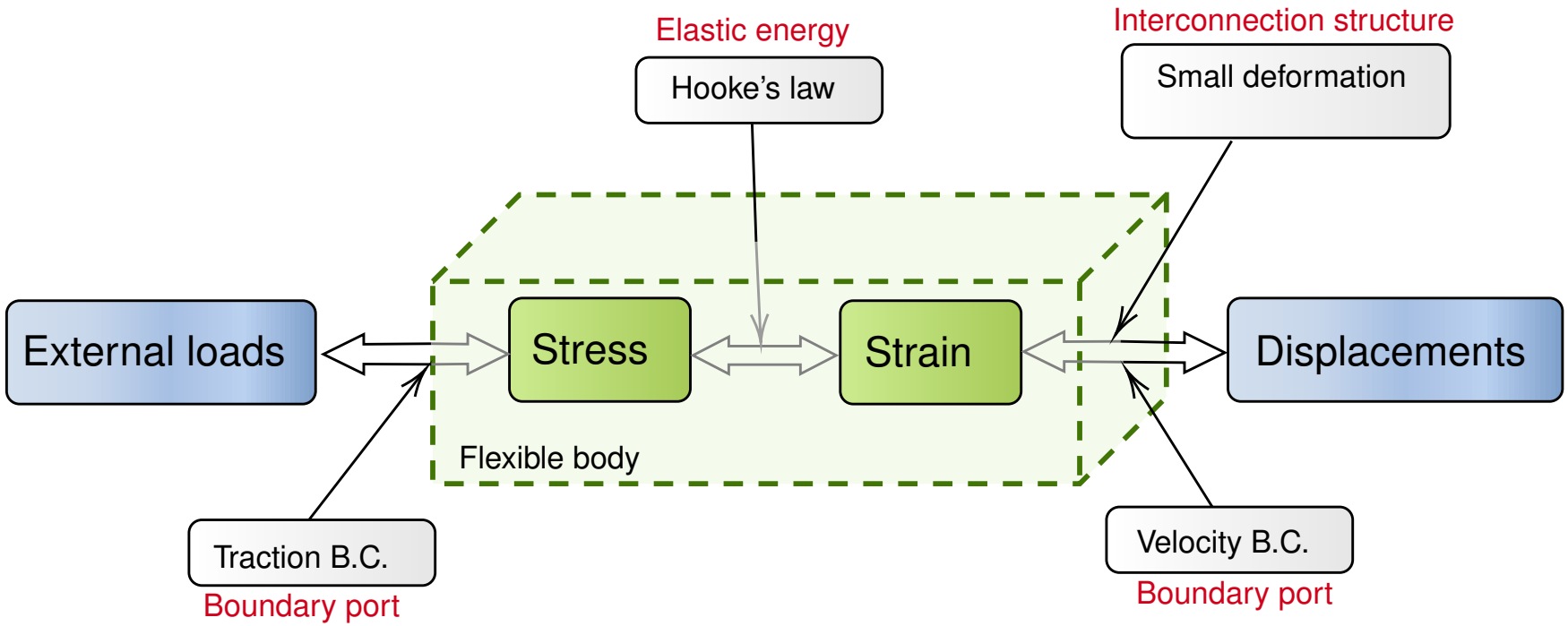


- A solid that undergoes continuous deformation under applied loads.

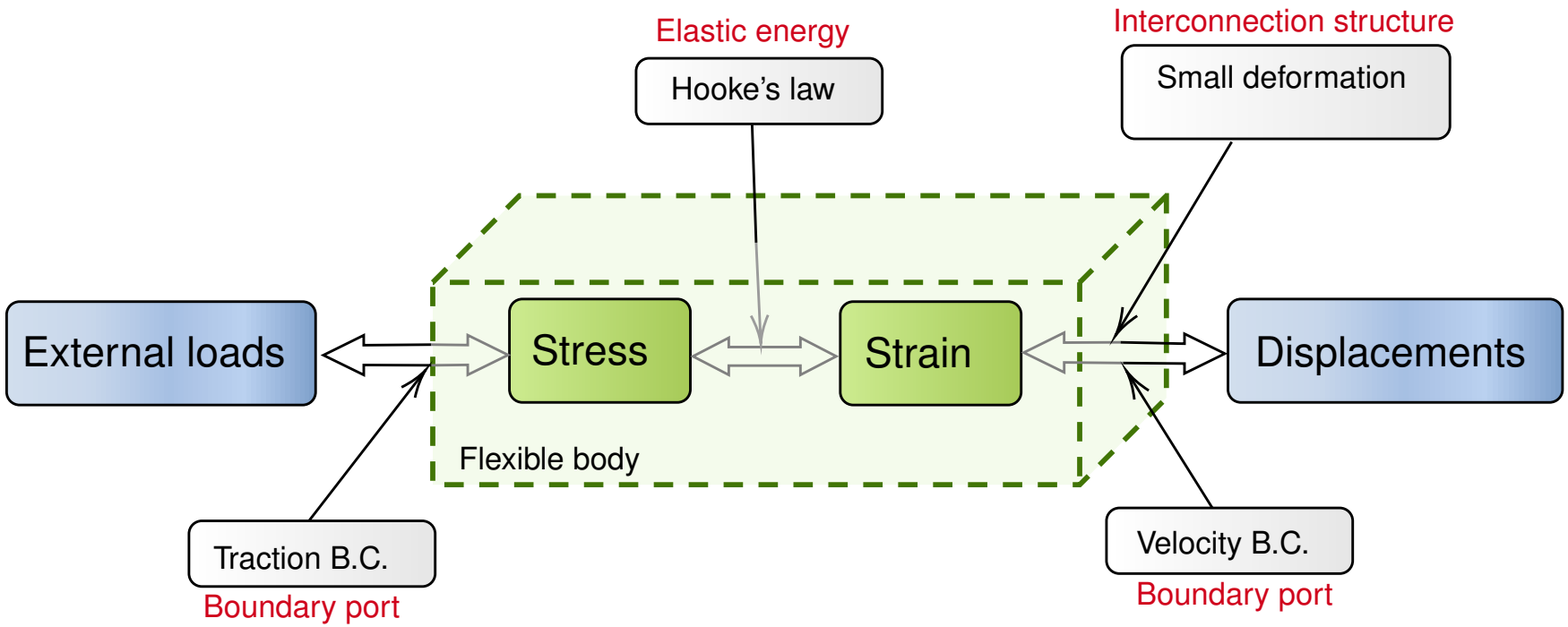
Background: Elastodynamics



Background: Linear Elastodynamics



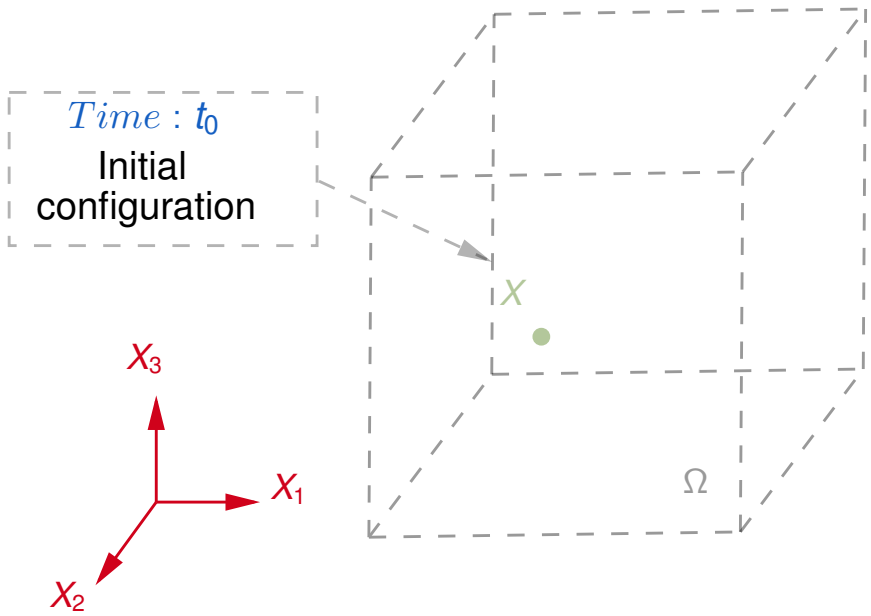
Background: Linear Elastodynamics



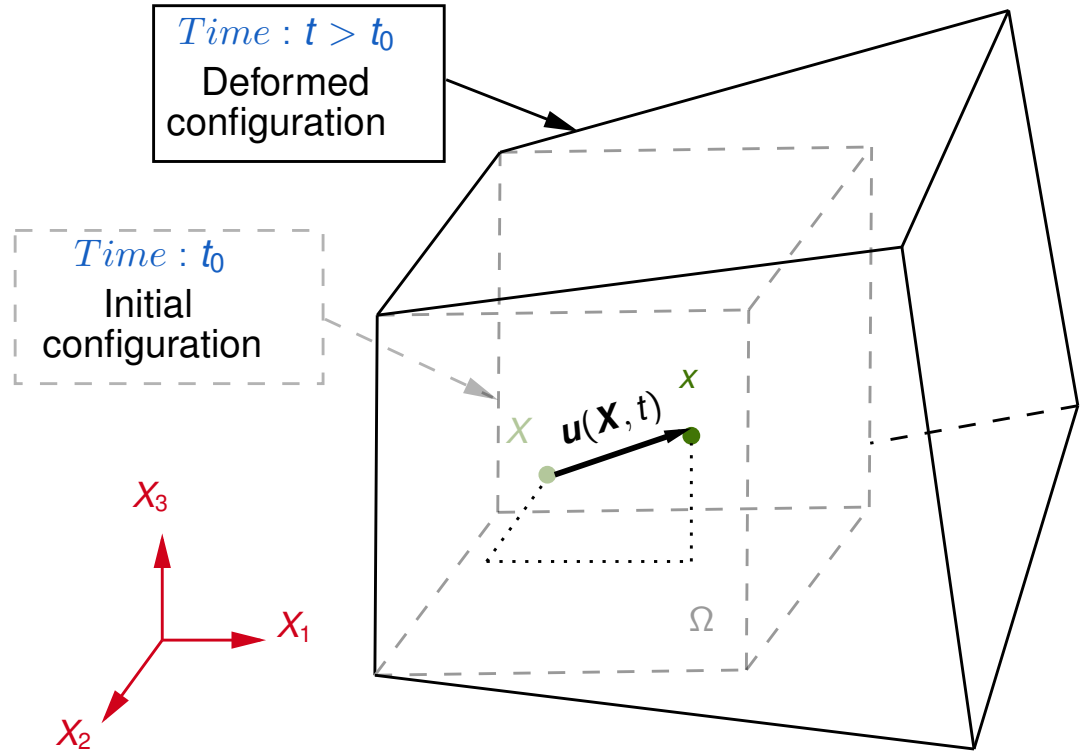
Main variables:

- Displacement: $u(X,t)$
- Strain: $\underline{\varepsilon}(X,t) = \frac{1}{2}(\nabla u + \nabla u^\top) = Grad(u)$ (small deformation)
- Stress: $\underline{\sigma}(X,t) = \underline{\underline{C}} : \underline{\varepsilon}(X,t)$ (Hooke's law)

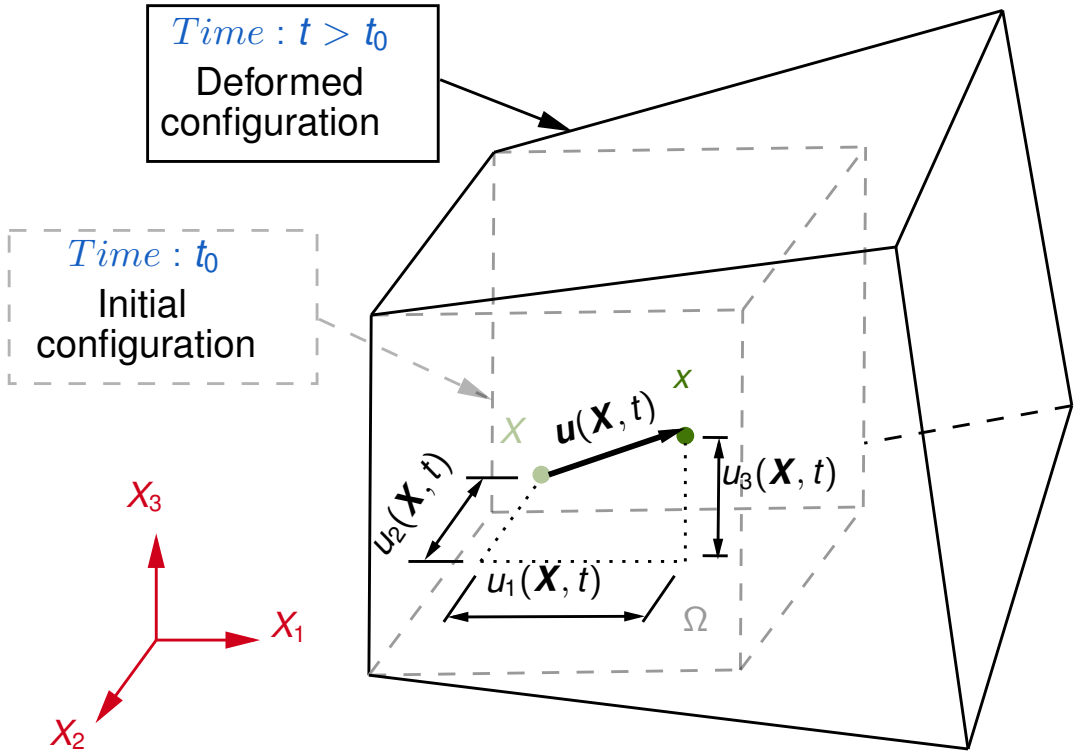
Background: Displacement field (3D case)



Background: Displacement field (3D case)



Background: Displacement field (3D case)



Background: Energies

Kinetic energy:

$$T = \frac{1}{2} \int_{\mathcal{B}_0} \rho(X) \dot{u}(X,t)^\top \dot{u}(X,t)\, dX$$

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External work:

$$W_E = \int_{\partial \mathcal{B}_0^N} u(S,t)^\top t_N(S,t)\, dS.$$

Hamilton's principle

$$\int_{t_1}^{t_2} [\delta(T - U) + \delta W_E]\, dt = 0, \quad \forall X \in \mathcal{B}_0 \cup \partial\mathcal{B}_0 \tag{7}$$

$$\delta u(X, t_1) = \delta u(X, t_2) = 0, \quad \forall X \in \mathcal{B}_0 \tag{8}$$

$$\delta u(S, t) = 0, \quad \forall S \in \partial\mathcal{B}_0^D \tag{9}$$

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Model:

$$\forall X \in \mathcal{B}_0 : \quad \rho\ddot{u} - \text{Div}\left(\underline{\underline{C}} : \text{Grad}(u)\right) = 0 \quad \text{(PDE)}$$

$$\forall S \in \partial\mathcal{B}_0^N : \quad \underline{\underline{C}} : \text{Grad}(u)\, \hat{n} = t_N \quad \text{(Neumann B.C.)}$$

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Evaluating term by term:

$$\rightarrow \int_{t_1}^{t_2} \delta T \, dt$$

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Hamilton's principle: Proof

Replacing the terms:

$$\int_{t_1}^{t_2}\int_{\mathcal{B}_0}\delta u^\top\Big[-\rho\ddot{u}+\mathrm{Div}\left(\underline{\underline{C}}:\mathrm{Grad}(u)\right)\Big]\,dX\,dt+\int_{t_1}^{t_2}\int_{\partial\mathcal{B}_0^N}\delta u^\top\Big[t_N-\underline{\underline{C}}:\mathrm{Grad}(u)\,\hat{n}\Big]\,dS\,dt=0$$

Hamilton's principle: Proof

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Since δu is arbitrary, then we obtain:

$$\forall X \in \mathcal{B}_0 : \quad \rho\ddot{u} - \mathrm{Div}\left(\underline{\underline{C}} : \mathrm{Grad}(u)\right) = 0 \qquad \text{(PDE)}$$

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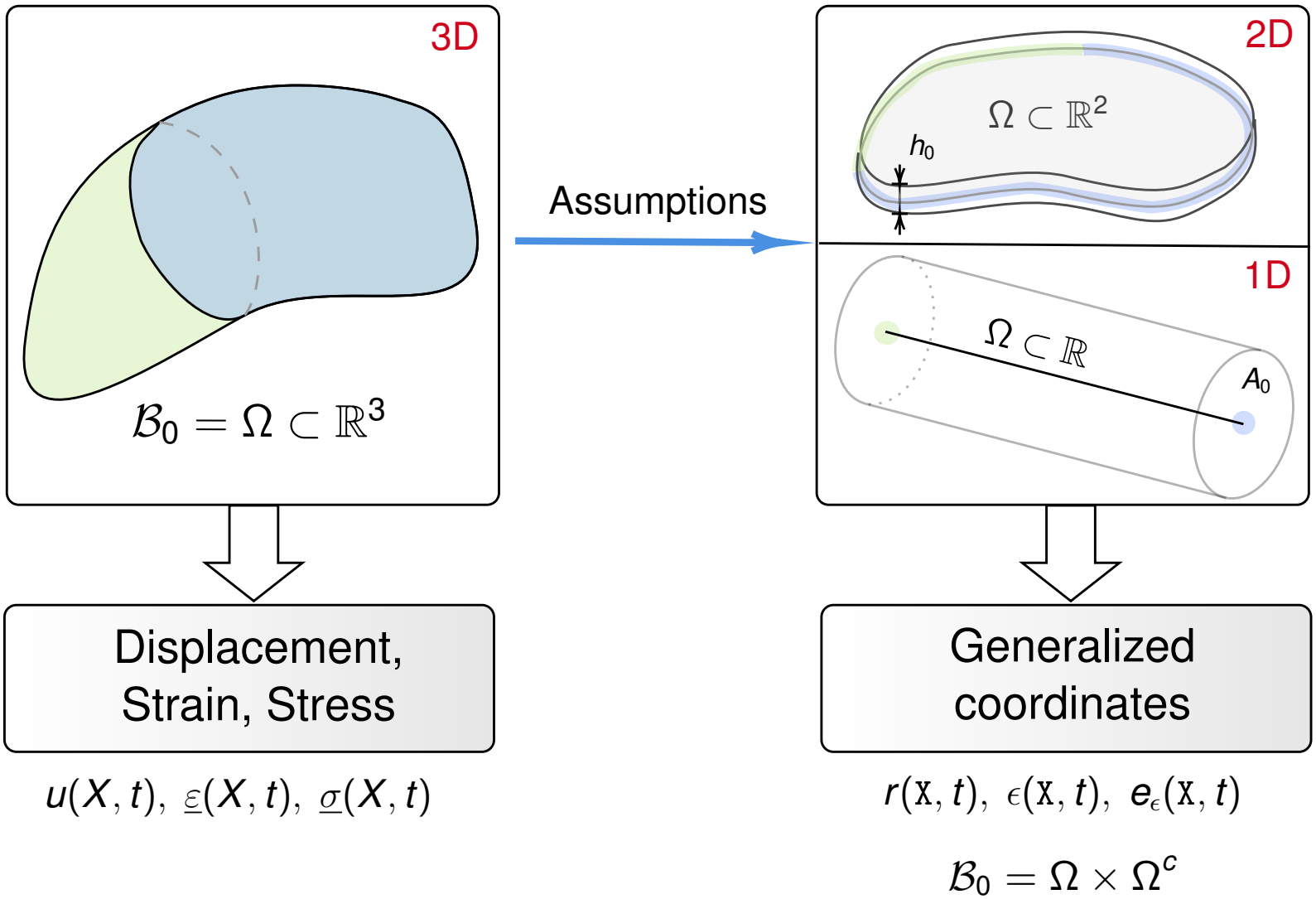
$$\int_{t_1}^{t_2}\int_{\mathcal{B}_0} \delta u^\top \underbrace{\left[-\rho\ddot{u} + \mathrm{Div}\left(\underline{\underline{C}} : \mathrm{Grad}(u)\right)\right]}_{0} dX\, dt + \int_{t_1}^{t_2}\int_{\partial\mathcal{B}_0^N} \delta u^\top \underbrace{\left[t_N - \underline{\underline{C}} : \mathrm{Grad}(u)\,\hat{n}\right]}_{0} dS\, dt = 0$$

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$$\begin{aligned}
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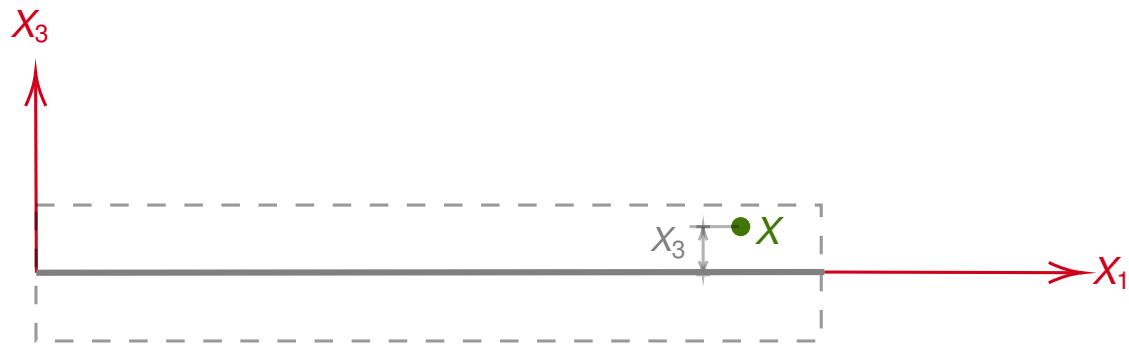
- **Keep in mind:** The integral equation above (from Hamilton's principle) is the starting point for the Finite element discretization.

Modeling: General idea



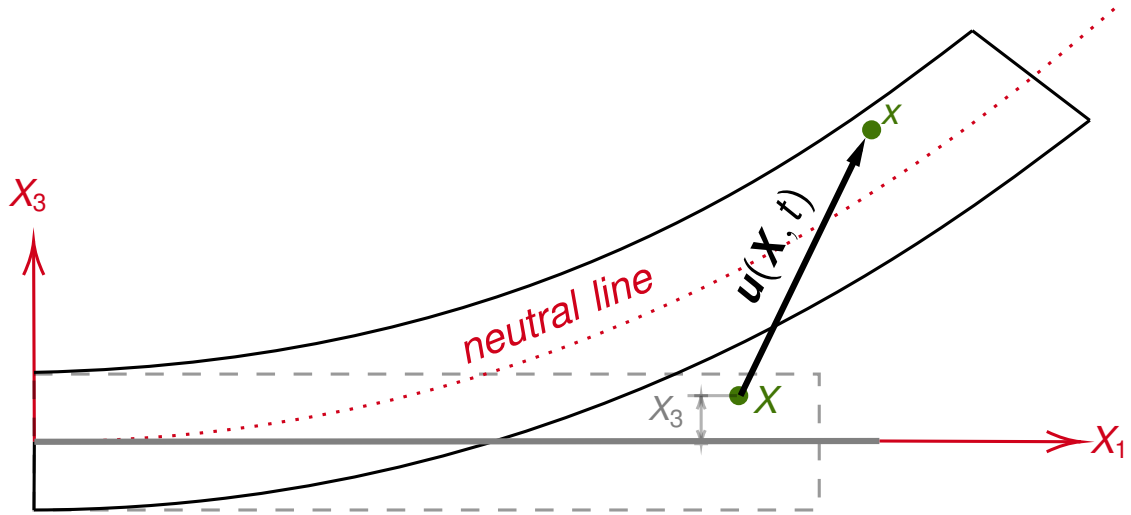
Example: Planar Timoshenko beam

Kinematic assumption: Plane sections normal to the neutral line before deformation remain plane but not necessarily normal to the neutral line after deformation.



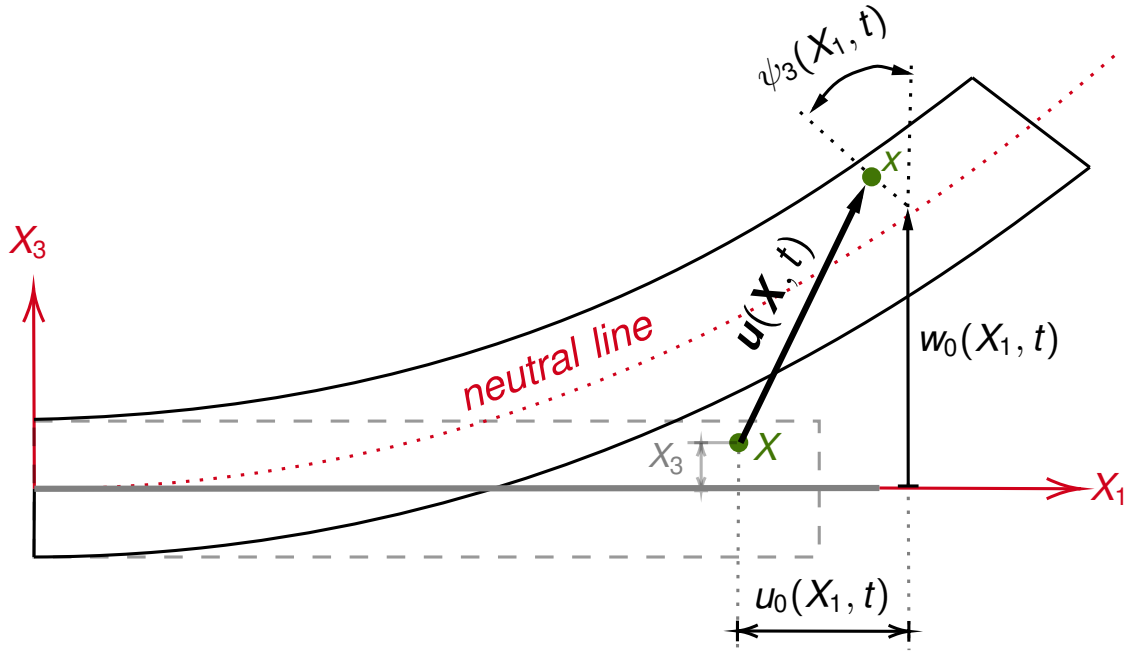
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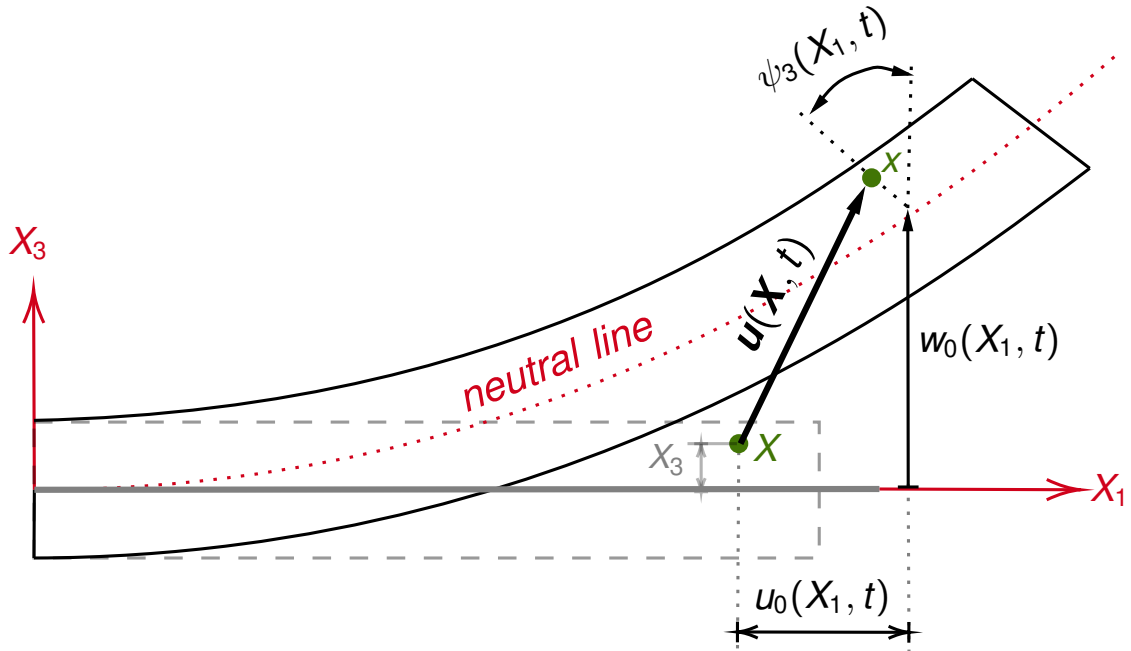
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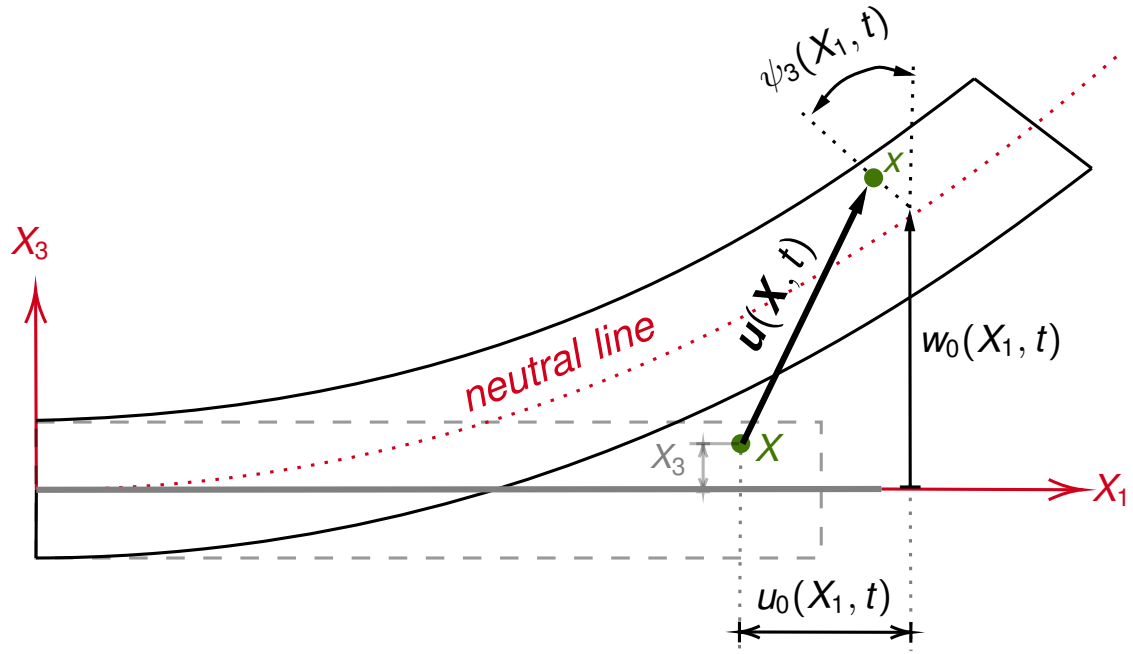
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$$\boldsymbol{u}(\boldsymbol{X}, t) = \underbrace{\begin{bmatrix} 1 & -X_3 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\overline{M}_1(\mathrm{x}^c)} \underbrace{\begin{bmatrix} u_0(\mathrm{x}, t) \\ \psi_3(\mathrm{x}, t) \\ w_0(\mathrm{x}, t) \end{bmatrix}}_{r(\mathrm{x},t)}$$

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Modeling: General idea

3D model:

$$\forall X \in \mathcal{B}_0 : \qquad \rho\ddot{u} - \mathrm{Div}\left(\underline{\underline{C}} : \mathrm{Grad}(u)\right) = 0 \qquad \text{(PDE)}$$

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Generalized model:

$$\forall \mathtt{X} \in \Omega : \qquad \mathcal{M}\ddot{r} + \mathcal{F}^*(\mathcal{K}_\epsilon \mathcal{F} r) = 0 \qquad \text{(PDE)}$$

$$\forall \mathtt{S} \in \partial\Omega_N : \qquad F_\partial(\hat{n})\,\mathcal{K}_\epsilon \mathcal{F} r = \tau_N \qquad \text{(Neumann B.C.)}$$

$$\forall \mathtt{S} \in \partial\Omega_D : \qquad r = r_D \qquad \text{(Dirichlet B.C.)}$$

Generalized linear elastic model

$$\forall \mathtt{x} \in \Omega : \quad \mathcal{M}\ddot{r} + \mathcal{F}^*(\mathcal{K}_\epsilon \mathcal{F} r) = 0 \qquad \text{(PDE)}$$

$$\forall \mathtt{s} \in \partial\Omega_N : \quad F_\partial(\hat{n})\, \mathcal{K}_\epsilon \mathcal{F} r = \tau_N \qquad \text{(Neumann B.C.)}$$

$$\forall \mathtt{s} \in \partial\Omega_D : \quad r = r_D \qquad \text{(Dirichlet B.C.)}$$

where:

- $r(\mathtt{x}, t)$: Generalized displacement vector.
- $\mathcal{M}(\mathtt{x})$: Mass density matrix.
- $\mathcal{K}_\epsilon(\mathtt{x})$: Strain-based stiffness density matrix.
- $\mathcal{F}$, $\mathcal{F}^*$: Linear differential operators.
- $F_\partial(\hat{n})$: Boundary-valued matrix operator induced by $\mathcal{F}$.

Other useful definitions:

- $\epsilon(\mathtt{x}, t) = \mathcal{F} r(\mathtt{x}, t)$ Generalized strain (kinematic equation).
- $e_\epsilon(\epsilon) = \mathcal{K}_\epsilon(\mathtt{x})\epsilon(\mathtt{x}, t)$ Generalized stress (constitutive equation).
- $\mathcal{K}_r(\cdot) = \mathcal{F}^*(\mathcal{K}_\epsilon(\mathtt{x})\mathcal{F}(\cdot))$ Displacement-based stiffness operator.

Generalized linear elastic PHS model (1)

With the definition of the generalized momentum $p(\mathtt{x},t) = \mathcal{M}(\mathtt{x})\dot{r}(\mathtt{x},t)$, the system defines the following PHS representation:

$$\underbrace{\begin{bmatrix}\dot{p}(\mathtt{x},t)\\ \dot{r}(\mathtt{x},t)\end{bmatrix}}_{\dot{z}(\mathtt{x},t)} = \underbrace{\begin{bmatrix}0 & -I\\ I & 0\end{bmatrix}}_{J=-J^\top} \underbrace{\begin{bmatrix}e_p(p)\\ e_r(r)\end{bmatrix}}_{\delta_z H(z)}$$

[1]

$$H(z) = \int_\Omega \left[\frac{1}{2}p(\mathtt{x},t)^\top \mathcal{M}(\mathtt{x})^{-1}p(\mathtt{x},t) + \frac{1}{2}(\mathcal{F}r(\mathtt{x},t))^\top \mathcal{K}_\epsilon(\mathtt{x})(\mathcal{F}r(\mathtt{x},t))\right] d\mathtt{x}$$

$$\dot{H} = \int_{\partial\Omega} \underbrace{e_p(p)^\top F_\partial(\hat{n})\, e_\epsilon(\epsilon(r))}_{y_\partial^\top u_\partial}\, dS$$

where $e_p(p) = \mathcal{M}(\mathtt{x})^{-1}p(\mathtt{x},t)$ and $e_r(r) = \mathcal{K}_r(r)$ are the generalized velocities and internal loads, respectively.

[1] $\delta_z H(z)$ denotes the variational derivatives of $H(z)$ with respect to $z(\mathtt{x},t)$.

Generalized linear elastic PHS model (2)

With the definition of the generalized momentum $p(\mathtt{X}, t) = \mathcal{M}(\mathtt{X})\dot{r}(\mathtt{X}, t)$ and the generalized strain $\epsilon(\mathtt{X}, t) = \mathcal{F} r(\mathtt{X}, t)$, the system defines the following PHS representation:

$$\underbrace{\begin{bmatrix} \dot{p}(\mathtt{X}, t) \\ \dot{\epsilon}(\mathtt{X}, t) \\ \dot{r}(\mathtt{X}, t) \end{bmatrix}}_{\dot{x}(\mathtt{X}, t)} = \underbrace{\begin{bmatrix} 0 & -\mathcal{F}^* & -I \\ \mathcal{F} & 0 & 0 \\ I & 0 & 0 \end{bmatrix}}_{\mathcal{J} = -\mathcal{J}^*} \underbrace{\begin{bmatrix} e_p(p) \\ e_\epsilon(\epsilon) \\ 0 \end{bmatrix}}_{\delta_x H(x)^2}$$

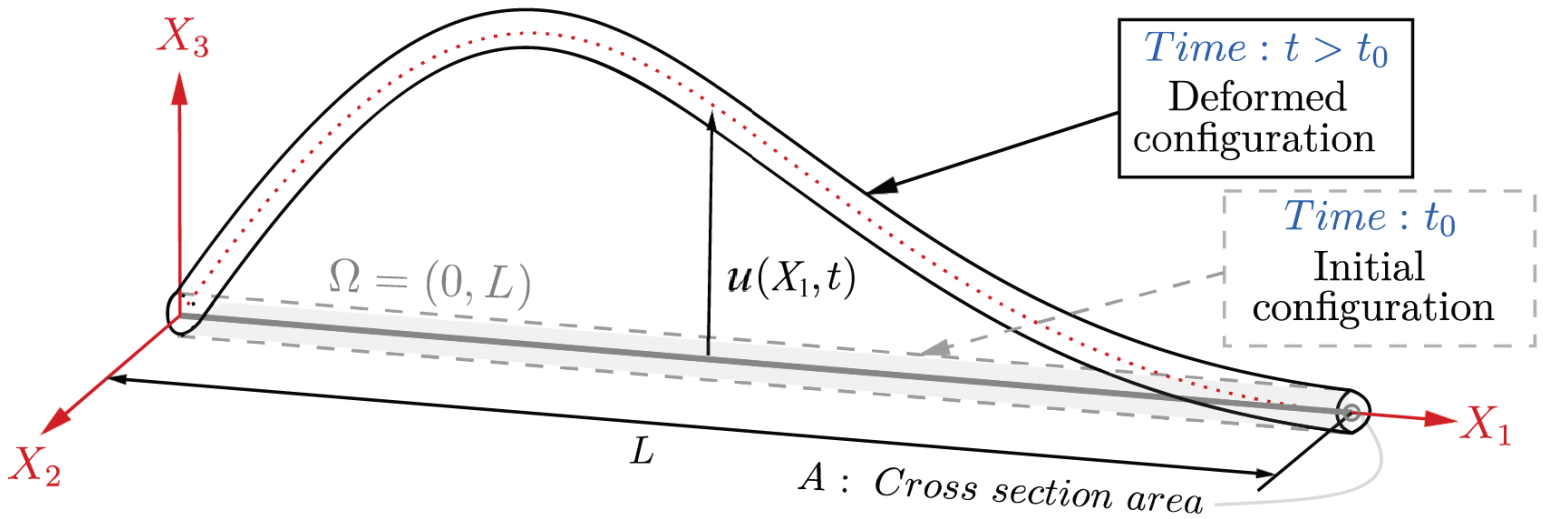
$$H(x) = \int_\Omega \left[\frac{1}{2} p(\mathtt{X}, t)^\top \mathcal{M}(\mathtt{X})^{-1} p(\mathtt{X}, t) + \frac{1}{2} \epsilon(\mathtt{X}, t)^\top \mathcal{K}_\epsilon(\mathtt{X}) \epsilon(\mathtt{X}, t) \right] d\mathtt{X}$$

$$\dot{H} = \int_{\partial\Omega} \underbrace{e_p(p)^\top F_\partial(\hat{n})\, e_\epsilon(\epsilon)}_{y_\partial^\top u_\partial} \, d\mathtt{S}$$

where $e_p(p) = \mathcal{M}(\mathtt{X})^{-1} p(\mathtt{X}, t) = \dot{r}(\mathtt{X}, t)$ and $e_\epsilon(\epsilon) = \mathcal{K}_\epsilon(\mathtt{X}, t)\epsilon(\mathtt{X}, t)$ are the generalized velocities and stresses, respectively.

[2] $\delta_x H(x)$ denotes the variational derivatives of $H(x)$ with respect to $x(\mathtt{X}, t)$.

Example: 1D vibrating string



Displacement field:

$$\mathbf{u}(\mathbf{X}, t) = \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}_{\bar{M}_1(\mathtt{x}^c)} \underbrace{\left[u(\mathtt{x}, t)\right]}_{r(\mathtt{x},t)} \quad , \quad \mathtt{x} = X_1$$

Example: 1D vibrating string

$$\rho A\ddot{u} - \frac{\partial}{\partial x_1}\left(T\,\frac{\partial u}{\partial x_1}\right) = 0$$

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- **PHS model (1):**

$$\underbrace{\begin{bmatrix} \dot{p} \\ \dot{u} \end{bmatrix}}_{\dot{z}} = \underbrace{\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}}_{J=-J^\top} \underbrace{\begin{bmatrix} p/\rho A \\ -\partial_1(T\partial_1 u) \end{bmatrix}}_{\delta_z H(z)}$$

$$H(z) = \int_0^L \left[\frac{1}{2}\frac{p^2}{\rho A} + \frac{1}{2} T (\partial_1 u)^2\right] dx_1$$

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- **PHS model (2):**

$$\underbrace{\begin{bmatrix} \dot{p} \\ \dot{\epsilon} \\ \dot{u} \end{bmatrix}}_{\dot{x}} = \underbrace{\begin{bmatrix} 0 & \partial_1 & -1 \\ \partial_1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}}_{\mathcal{J}=-\mathcal{J}^*} \underbrace{\begin{bmatrix} p/\rho A \\ T\epsilon \\ 0 \end{bmatrix}}_{\delta_x H(x)}$$

$$H(x) = \int_0^L \left[\frac{1}{2}\frac{p^2}{\rho A} + \frac{1}{2} T \epsilon^2\right] dx_1$$

Example: 1D vibrating string

$$\rho A\ddot{u} + d\dot{u} - \frac{\partial}{\partial x_1}\left(T\,\frac{\partial u}{\partial x_1}\right) = 0$$

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$$\underbrace{\begin{bmatrix}\dot{p}\\ \dot{u}\end{bmatrix}}_{\dot{z}} = \underbrace{\begin{bmatrix}-d & -1\\ 1 & 0\end{bmatrix}}_{(J-R)}\underbrace{\begin{bmatrix}p/\rho A\\ -\partial_1(T\partial_1 u)\end{bmatrix}}_{\delta_z H(z)}$$

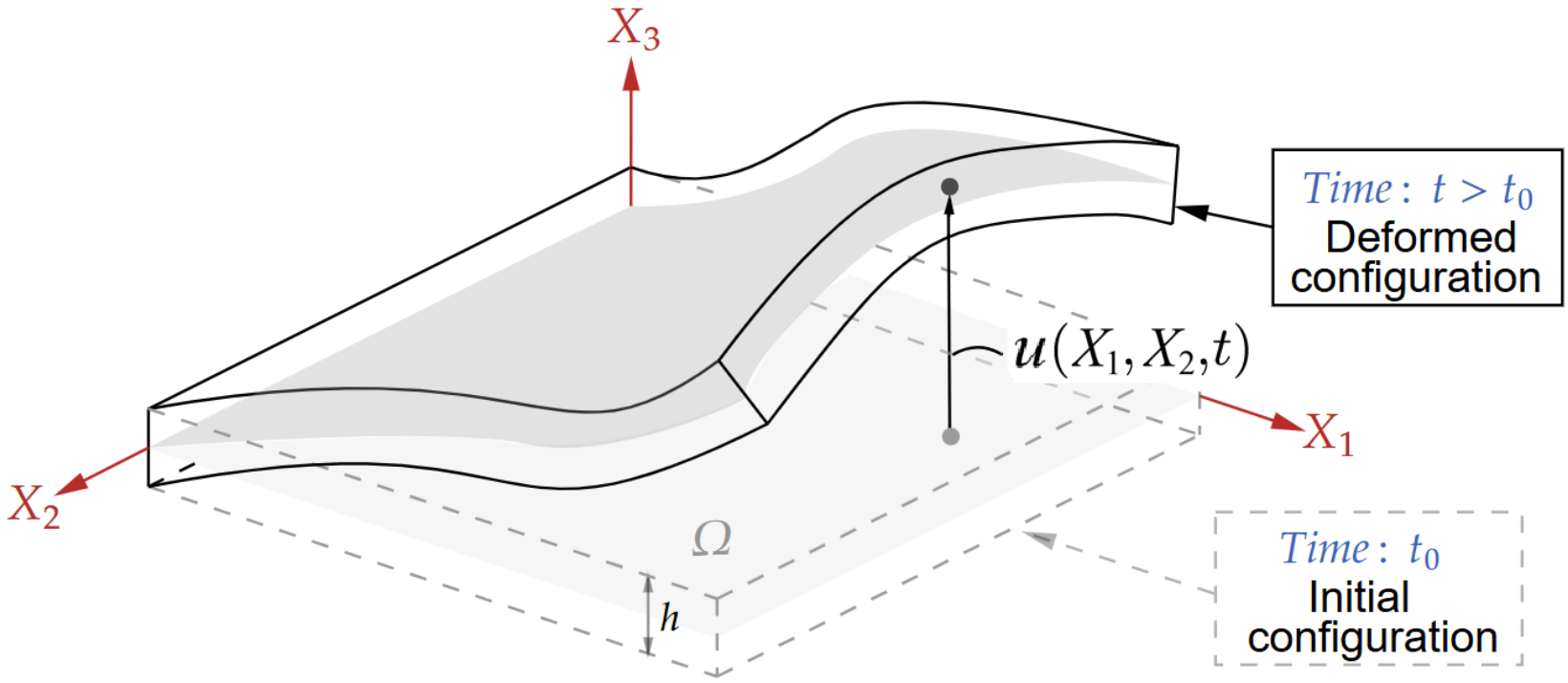
$$H(z) = \int_0^L \left[\frac{1}{2}\frac{p^2}{\rho A} + \frac{1}{2}T(\partial_1 u)^2\right] dx_1$$

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Example: 2D vibrating membrane



Displacement field:

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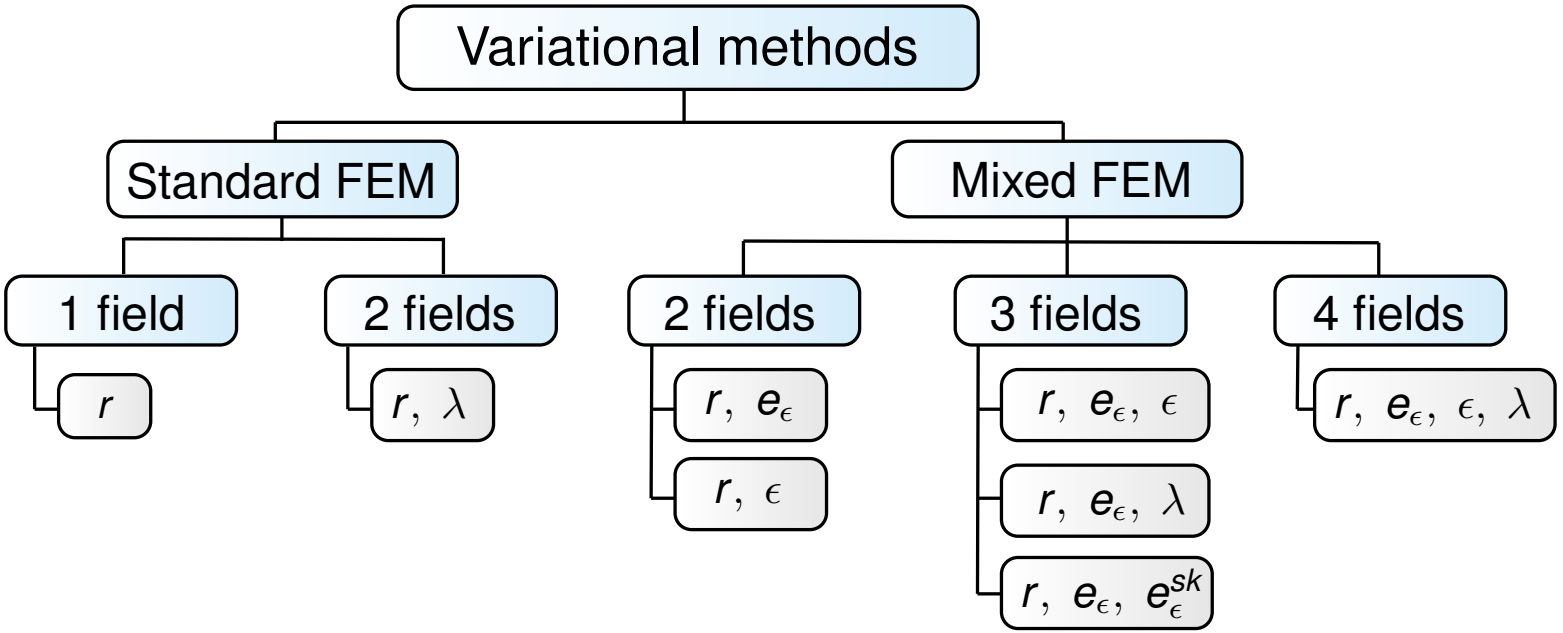
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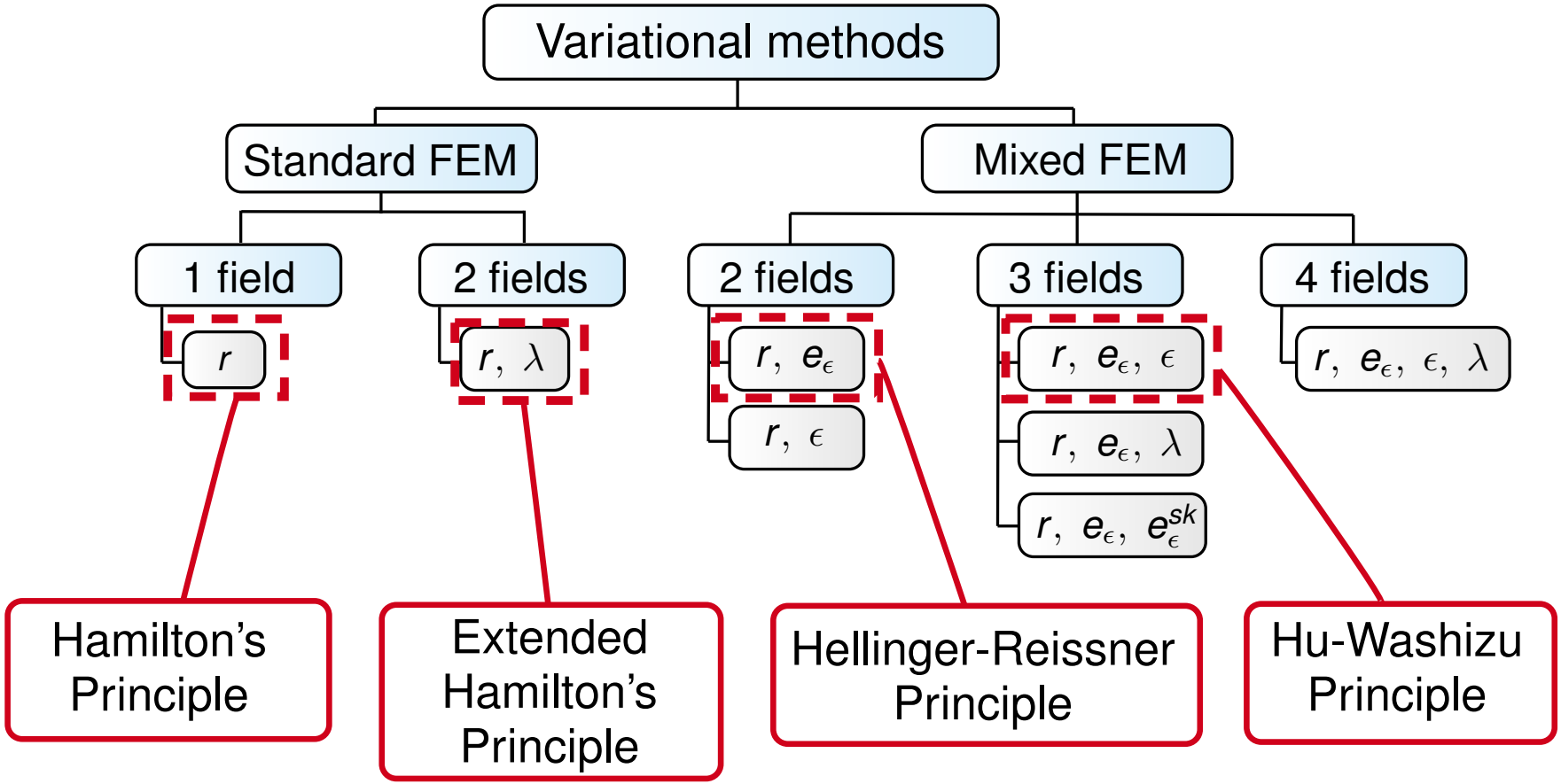
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3. Finite element discretizations

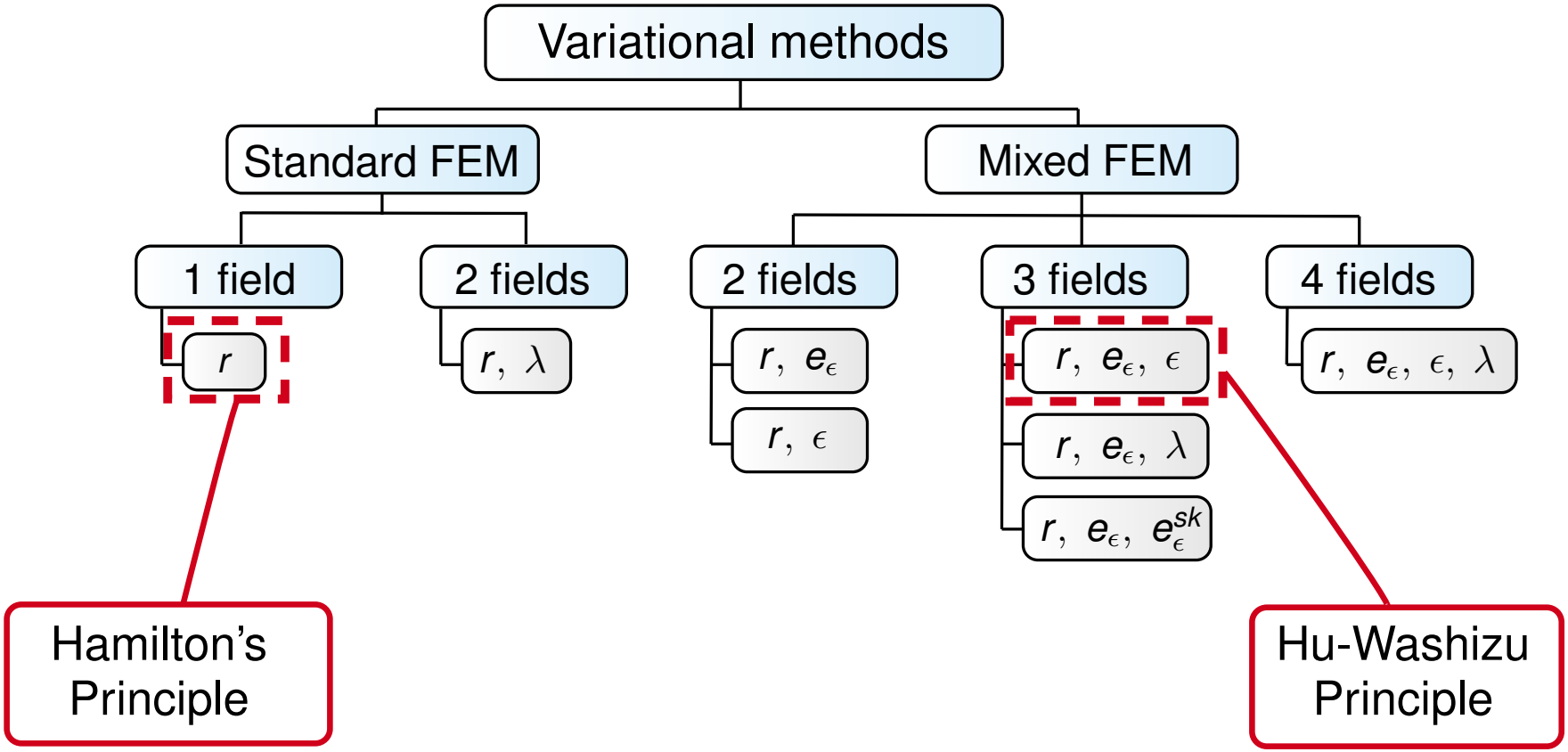
Finite element methods in solid mechanics



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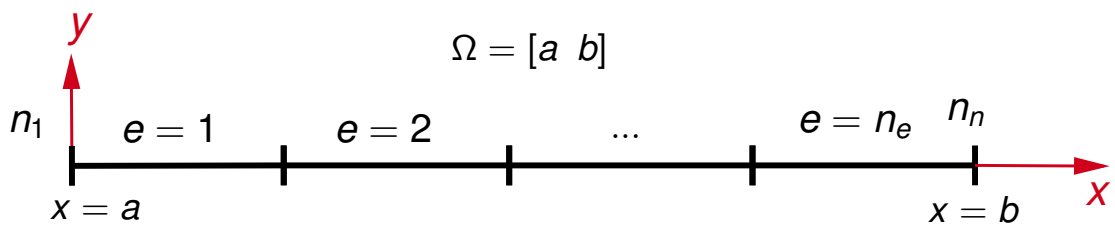
Finite element methods in solid mechanics



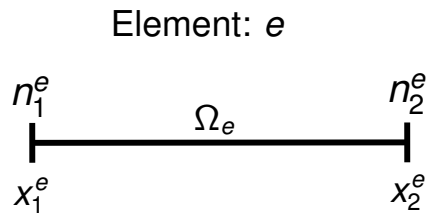
Preliminaries

Definitions:

- Finite elements: Finite and disjoint subdomains of the spatial domain.
- Nodes: Points in the domain that define the boundaries of each element.
- Mesh: The elements, nodes, and their connectivity topology.



a) Global domain



b) Local domain

Preliminaries

- **Local approximation**: The unknown $r(x,t)$ is approximated locally as:

$$\tilde{r}^e(x,t) = \alpha_0^e(t) + \alpha_1^e(t)x$$

where $\alpha_0^e(t)$ and $\alpha_1^e(t)$ are the unknown coefficients to be determined.

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- These coefficients are not convenient as the unknowns of the problem. Then, we rewrite the approximation using the following procedure:

$$\begin{aligned}\tilde{r}(x_1^e,t) &= \hat{r}_1^e(t) = \alpha_0^e(t) + \alpha_1^e(t)x_1^e \\ \tilde{r}(x_2^e,t) &= \hat{r}_2^e(t) = \alpha_0^e(t) + \alpha_1^e(t)x_2^e\end{aligned} \quad \Longleftrightarrow \quad \underbrace{\begin{bmatrix}\hat{r}_1^e(t)\\ \hat{r}_2^e(t)\end{bmatrix}}_{\hat{r}^e(t)} = \underbrace{\begin{bmatrix}1 & x_1^e\\ 1 & x_2^e\end{bmatrix}}_{M^e_{coef}} \underbrace{\begin{bmatrix}\alpha_0^e(t)\\ \alpha_1^e(t)\end{bmatrix}}_{\alpha^e(t)}$$

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- Solving for $\alpha^e(t)$ and substituting back, we obtain $\tilde{r}^e(x,t) = N^e(x)\,\hat{r}^e(t)$, where:

$$N^e(x) = \Big[\underbrace{\frac{x_2^e - x}{x_2^e - x_1^e}}_{N_1^e(x)} \quad \underbrace{\frac{x - x_1^e}{x_2^e - x_1^e}}_{N_2^e(x)}\Big] \quad \text{(Shape functions)}$$

Preliminaries

- **Local to global mapping**: We define the global vector of unknowns $\hat{r}(t) = [\hat{r}_1(t) \ \ldots \ \hat{r}_{n_n}(t)]^\top$. The local-to-global mapping is:

$$\hat{r}^e(t) = \hat{L}^e \hat{r}(t),$$

where $\hat{L}^e$ is the element assembly matrix.

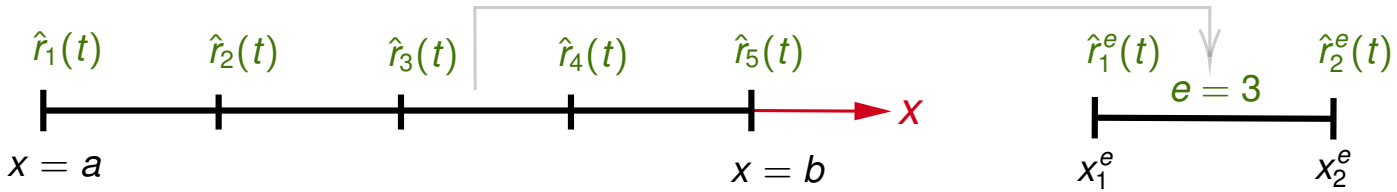
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Example: Consider the mesh in the figure and the finite element $e = 3$.



d) Local – to – global mapping

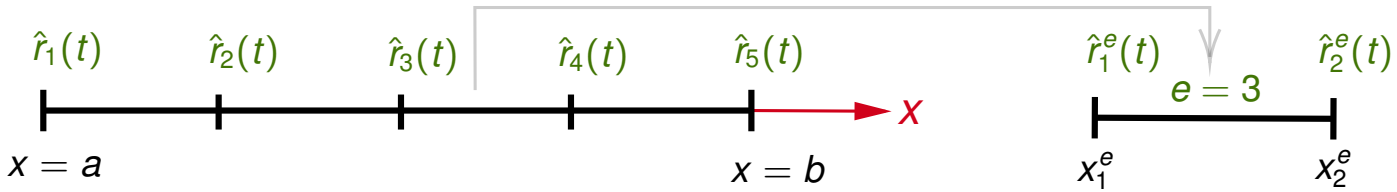
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d) Local – to – global mapping

The assembly matrix for element $e = 3$ is defined as:

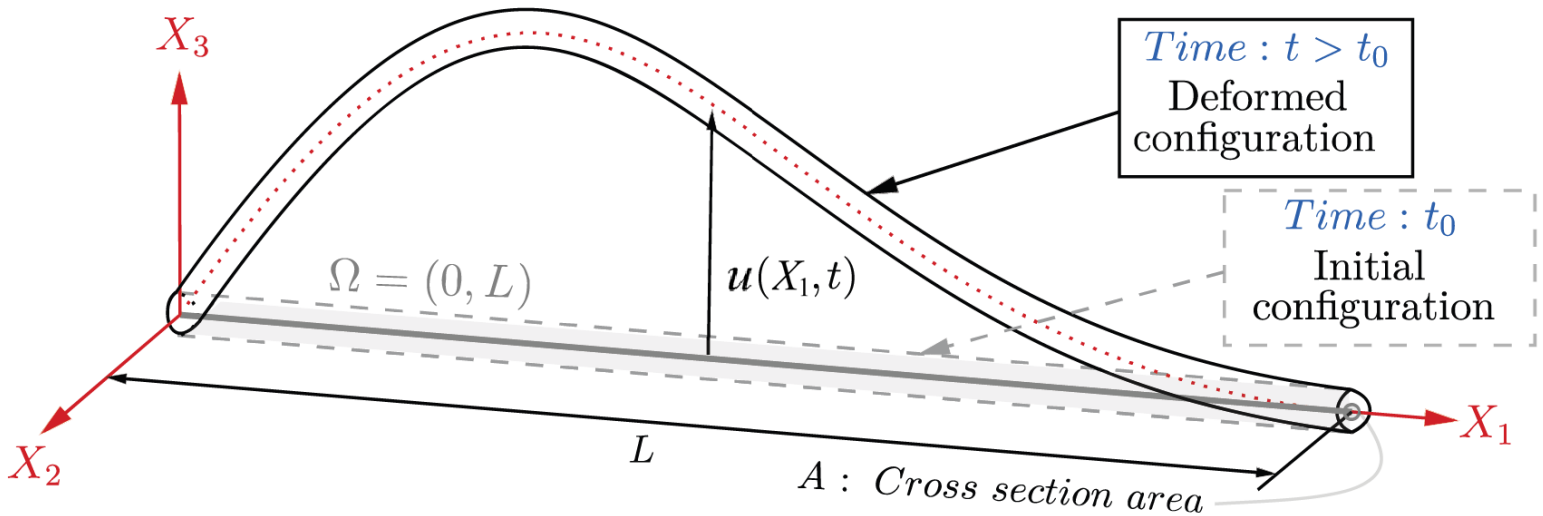
$$\underbrace{\begin{bmatrix} \hat{r}_1^e(t) \\ \hat{r}_2^e(t) \end{bmatrix}}_{\hat{r}^e(t)} = \underbrace{\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}}_{\hat{L}^e} \underbrace{\begin{bmatrix} \hat{r}_1(t) \\ \hat{r}_2(t) \\ \hat{r}_3(t) \\ \hat{r}_4(t) \\ \hat{r}_5(t) \end{bmatrix}}_{\hat{r}(t)} = \begin{bmatrix} \hat{r}_3(t) \\ \hat{r}_4(t) \end{bmatrix} \quad \text{(for } e = 3\text{)}.$$

Example: 1D vibrating string

$$\rho A\ddot{u} + d\dot{u} - \frac{\partial}{\partial x}\left(T\frac{\partial u}{\partial x}\right) = 0 \quad \text{(PDE)}$$

$$\dot{u}(x,t)\Big|_{x=0} = v_D(t) \quad \text{(Dirichlet B.C.)}$$

$$T\frac{\partial u}{\partial x}\Big|_{x=L} = F_L(t) \quad \text{(Neumann B.C.)}$$

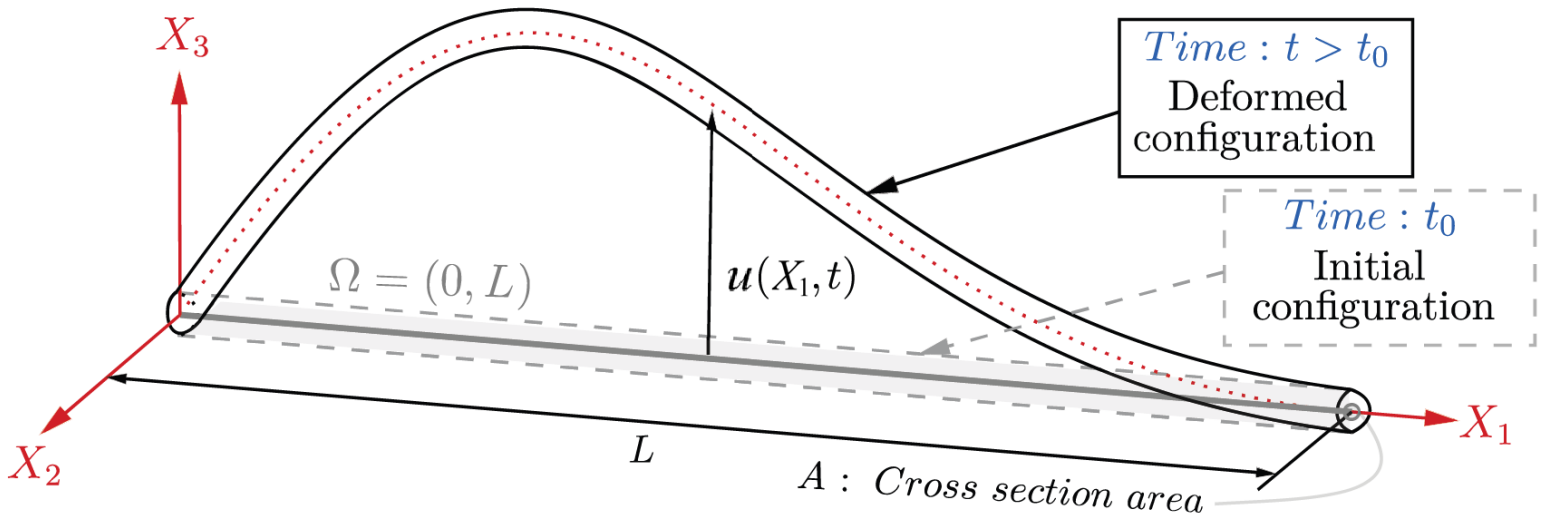


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- This will be done in **5 steps**.

Standard finite element method

1. The weak form:

From Hamilton's principle we obtain:

$$\delta\mathcal{S} = \int_{t_1}^{t_2} \left[\int_0^L \delta u^\top \left[\rho A \ddot{u} + d\dot{u} - \frac{\partial}{\partial x}\left(T \frac{\partial u}{\partial x} \right) \right] dx + \int_{\partial\Omega_N} \delta u^\top \left(T \frac{\partial u}{\partial x} \hat{n} - F_L \right) ds \right] dt = 0$$

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Expanding:

$$\delta\mathcal{S}=\int_{t_1}^{t_2}\left[\int_0^L\left[\delta u^\top\rho A\ddot{u}+\delta u^\top d\dot{u}-\delta u^\top\frac{\partial}{\partial x}\left(T\frac{\partial u}{\partial x}\right)\right]dx+\int_{\partial\Omega_N}\left(\delta u^\top T\frac{\partial u}{\partial x}\hat{n}-\delta u^\top F_L\right)ds\right]dt=0$$

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Expanding:

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$$\delta\mathcal{S} = \int_{t_1}^{t_2} \left[\int_0^L \delta u^\top \left[\rho A \ddot{u} + d\dot{u} - \frac{\partial}{\partial x}\left(T \frac{\partial u}{\partial x} \right) \right] dx + \int_{\partial\Omega_N} \delta u^\top \left(T \frac{\partial u}{\partial x} \hat{n} - F_L \right) ds \right] dt = 0$$

Expanding:

$$\delta\mathcal{S} = \int_{t_1}^{t_2} \left[\int_0^L \left[\delta u^\top \rho A \ddot{u} + \delta u^\top d\dot{u} - \underbrace{\delta u^\top \frac{\partial}{\partial x}\left(T \frac{\partial u}{\partial x} \right)}_{\text{i.b.p. (space)}} \right] dx + \int_{\partial\Omega_N} \left(\delta u^\top T \frac{\partial u}{\partial x} \hat{n} - \delta u^\top F_L \right) ds \right] dt = 0$$

After i.b.p. we obtain the following equation known as the weak form of the PDE:

$$\Pi = \int_0^L \left[\delta u^\top \rho A \ddot{u} + \delta u^\top d\dot{u} + \frac{\partial \delta u}{\partial x}^\top T \frac{\partial u}{\partial x} \right] dx - \int_{\partial\Omega_N} \delta u^\top F_L \, ds$$

Standard finite element method

2. Local weak form:

$$\Pi^e = \int_{x_1^e}^{x_2^e} \left[(\delta u^e)^\top \rho A \ddot{u}^e + (\delta u^e)^\top d \dot{u}^e + \frac{\partial \delta u^e}{\partial x}^\top T \frac{\partial u^e}{\partial x} \right] dx - \int_{\partial \Omega_N} (\delta u^e)^\top F_L \, ds$$

Standard finite element method

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3. Local matrices:

We have to replace $\tilde{u}^e(x,t) = N^e(x)\hat{u}^e(t)$ and $\delta\tilde{u}^e(x,t) = N^e(x)\delta\hat{u}^e(t)$ above:

Standard finite element method

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$$\Pi^e = \int_{x_1^e}^{x_2^e} \left[(\delta u^e)^\top \rho A \ddot{u}^e + (\delta u^e)^\top d \dot{u}^e + \frac{\partial \delta u^e}{\partial x}^\top T \frac{\partial u^e}{\partial x} \right] dx - \underbrace{\int_{\partial \Omega_N} (\delta u^e)^\top F_L \, ds}_{\neq 0 \text{ only in } e = n_n}$$

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We have to replace $\tilde{u}^e(x,t) = N^e(x)\hat{u}^e(t)$ and $\delta\tilde{u}^e(x,t) = N^e(x)\delta\hat{u}^e(t)$ above:

- $$\int_{x_1^e}^{x_2^e} (\delta u^e)^\top \rho A \ddot{u}^e dx = \delta\hat{u}^e(t)^\top \int_{x_1^e}^{x_2^e} N^e(x)^\top \rho A \, N^e(x) dx \, \ddot{\hat{u}}^e(t) = \delta\hat{u}^e(t)^\top \hat{M}^e \ddot{\hat{u}}^e(t)$$

Standard finite element method

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$$\Pi^e = \int_{x_1^e}^{x_2^e} \left[(\delta u^e)^\top \rho A \ddot{u}^e + (\delta u^e)^\top d \dot{u}^e + \frac{\partial \delta u^e}{\partial x}^\top T \frac{\partial u^e}{\partial x} \right] dx - \underbrace{\int_{\partial \Omega_N} (\delta u^e)^\top F_L \, ds}_{\neq 0 \text{ only in } e = n_n}$$

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- $$\int_{x_1^e}^{x_2^e} (\delta u^e)^\top d \, \dot{u}^e dx = \delta\hat{u}^e(t)^\top \int_{x_1^e}^{x_2^e} N^e(x)^\top d \, N^e(x) dx \, \dot{\hat{u}}^e(t) = \delta\hat{u}^e(t)^\top \hat{D}^e \dot{\hat{u}}^e(t)$$

Standard finite element method

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- $$\int_{x_1^e}^{x_2^e} (\delta u^e)^\top d \, \dot{u}^e dx = \delta\hat{u}^e(t)^\top \int_{x_1^e}^{x_2^e} N^e(x)^\top d \, N^e(x) dx \, \dot{\hat{u}}^e(t) = \delta\hat{u}^e(t)^\top \hat{D}^e \dot{\hat{u}}^e(t)$$
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Standard finite element method

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- $$\int_{x_1^e}^{x_2^e} (\delta u^e)^\top d \, \dot{u}^e dx = \delta\hat{u}^e(t)^\top \int_{x_1^e}^{x_2^e} N^e(x)^\top d \, N^e(x) dx \, \dot{\hat{u}}^e(t) = \delta\hat{u}^e(t)^\top \hat{D}^e \dot{\hat{u}}^e(t)$$
- $$\int_{x_1^e}^{x_2^e} \frac{\partial \delta u^e}{\partial x}^\top T \frac{\partial u^e}{\partial x} dx = \delta\hat{u}^e(t)^\top \int_{x_1^e}^{x_2^e} \frac{\partial N^e}{\partial x}^\top T \frac{\partial N^e}{\partial x} dx \, \hat{u}^e(t) = \delta\hat{u}^e(t)^\top \hat{K}_r^e \hat{u}^e(t)$$
- $$\int_{\partial\Omega_N} (\delta u^e)^\top F_L \, ds = \delta\hat{u}^e(t)^\top \int_{\partial\Omega_N} N^e(s)^\top ds \, F_L(t) = \delta\hat{u}^e(t)^\top \hat{B}^e F_L(t)$$

Standard finite element method

4. Assembly: $\Pi = \sum_{e=1}^{n_e} \Pi^e$

Standard finite element method

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Substituting the approximations into the local weak form yields:

$$\Pi^e = \delta\hat{u}^e(t)^\top \left[\hat{M}^e \ddot{\hat{u}}^e(t) + \hat{D}^e \dot{\hat{u}}^e(t) + \hat{K}_r^e \hat{u}^e(t) - \hat{B}^e F_L(t) \right]$$

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Introducing the global vectors:

$$\hat{u}(t) = \begin{bmatrix} \hat{u}_1(t) & \hat{u}_2(t) & \dots & \hat{u}_{n_n}(t) \end{bmatrix}^\top$$
$$\delta\hat{u}(t) = \begin{bmatrix} \delta\hat{u}_1(t) & \delta\hat{u}_2(t) & \dots & \delta\hat{u}_{n_n}(t) \end{bmatrix}^\top$$

and applying the local-to-global mappings $\hat{u}^e(t) = \hat{L}^e\,\hat{u}(t), \quad \delta\hat{u}^e(t) = \hat{L}^e\,\delta\hat{u}(t)$, the global weak form becomes:

$$\Pi = \sum_{e=1}^{n_e} \Pi^e = \delta\hat{u}(t)^\top \sum_{e=1}^{n_e} (\hat{L}^e)^\top \left[\hat{M}^e \ddot{\hat{u}}^e(t) + \hat{D}^e \dot{\hat{u}}^e(t) + \hat{K}_r^e \hat{u}^e(t) - \hat{B}^e F_L(t) \right]$$

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$$= \delta\hat{u}(t)^\top \sum_{e=1}^{n_e} (\hat{L}^e)^\top \left[\hat{M}^e \hat{L}^e \ddot{\hat{u}}(t) + \hat{D}^e \hat{L}^e \dot{\hat{u}}(t) + \hat{K}_r^e \hat{L}^e \hat{u}(t) - \hat{B}^e F_L(t) \right]$$

Standard finite element method

Defining the global matrices as:

$$\hat{M} = \sum_{e=1}^{n_e} (\hat{L}^e)^\top \hat{M}^e \hat{L}^e, \quad \hat{D} = \sum_{e=1}^{n_e} (\hat{L}^e)^\top \hat{D}^e \hat{L}^e, \quad \hat{K}_r = \sum_{e=1}^{n_e} (\hat{L}^e)^\top \hat{K}_r^e \hat{L}^e, \quad \hat{B} = \sum_{e=1}^{n_e} (\hat{L}^e)^\top \hat{B}^e,$$

the global weak form is finally expressed as:

$$\Pi = \sum_{e=1}^{n_e} \Pi^e = \delta\hat{u}(t)^\top \left[\hat{M}\ddot{\hat{u}}(t) + \hat{D}\dot{\hat{u}}(t) + \hat{K}_r \hat{u}(t) - \hat{B} F_L(t) \right]$$

Standard finite element method

Defining the global matrices as:

$$\hat{M}=\sum_{e=1}^{n_e}(\hat{L}^e)^\top\hat{M}^e\hat{L}^e,\quad \hat{D}=\sum_{e=1}^{n_e}(\hat{L}^e)^\top\hat{D}^e\hat{L}^e,\quad \hat{K}_r=\sum_{e=1}^{n_e}(\hat{L}^e)^\top\hat{K}_r^e\hat{L}^e,\quad \hat{B}=\sum_{e=1}^{n_e}(\hat{L}^e)^\top\hat{B}^e,$$

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5. The PHS model:

Since $\delta\mathcal{S}=\int_{t_1}^{t_2}\Pi\,dt=0$ holds for any arbitrary $\delta\hat{u}(t)$, we obtain:

$$\hat{M}\ddot{\hat{u}}(t)+\hat{D}\dot{\hat{u}}(t)+\hat{K}_r\hat{u}(t)=\hat{B}F_L(t)$$

Standard finite element method

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5. The PHS model:

Since $\delta\mathcal{S}=\int_{t_1}^{t_2}\Pi\,dt=0$ holds for any arbitrary $\delta\hat{u}(t)$, we obtain:

$$\hat{M}\ddot{\hat{u}}(t)+\hat{D}\dot{\hat{u}}(t)+\hat{K}_r\hat{u}(t)=\hat{B}F_L(t)\qquad\text{(ODE + Neumann B.C.)}$$

How do we impose Dirichlet B.C.?

Standard finite element method

Consider the partition:

$$\hat{u}(t) = \left[\hat{u}_1(t) \mid \hat{u}_2(t) \quad \dots \quad \hat{u}_{n_n}(t)\right]^\top$$

Standard finite element method

Consider the partition:

$$\hat{u}(t) = \begin{bmatrix} \hat{u}_1(t) & | & \hat{u}_2(t) & \dots & \hat{u}_{n_n}(t) \end{bmatrix}^\top = \begin{bmatrix} \hat{u}_D(t) & \hat{u}_\Omega(t) \end{bmatrix}^\top$$

Standard finite element method

Consider the partition:

$$\hat{u}(t) = \begin{bmatrix} \hat{u}_1(t) & | & \hat{u}_2(t) & \dots & \hat{u}_{n_n}(t) \end{bmatrix}^\top = \begin{bmatrix} \hat{u}_D(t) & \hat{u}_\Omega(t) \end{bmatrix}^\top$$

Then:

$$\underbrace{\begin{bmatrix} \hat{M}_{DD} & \hat{M}_{D\Omega} \\ \hat{M}_{\Omega D} & \hat{M}_{\Omega\Omega} \end{bmatrix}}_{\hat{M}} \underbrace{\begin{bmatrix} \ddot{\hat{u}}_D(t) \\ \ddot{\hat{u}}_\Omega(t) \end{bmatrix}}_{\ddot{\hat{u}}(t)} + \underbrace{\begin{bmatrix} \hat{D}_{DD} & \hat{D}_{D\Omega} \\ \hat{D}_{\Omega D} & \hat{D}_{\Omega\Omega} \end{bmatrix}}_{\hat{D}} \underbrace{\begin{bmatrix} \dot{\hat{u}}_D(t) \\ \dot{\hat{u}}_\Omega(t) \end{bmatrix}}_{\dot{\hat{u}}(t)} + \underbrace{\begin{bmatrix} \hat{K}_{DD} & \hat{K}_{D\Omega} \\ \hat{K}_{\Omega D} & \hat{K}_{\Omega\Omega} \end{bmatrix}}_{\hat{K}_r} \underbrace{\begin{bmatrix} \hat{u}_D(t) \\ \hat{u}_\Omega(t) \end{bmatrix}}_{\hat{u}(t)} = \underbrace{\begin{bmatrix} 0 \\ \hat{B}_N \end{bmatrix}}_{\hat{B}} F_L(t)$$

where $\dot{\hat{u}}_D(t) = v_D(t)$ is the Dirichlet B.C. to be imposed.

Standard finite element method

Consider the partition:

$$\hat{u}(t) = \begin{bmatrix} \hat{u}_1(t) & | & \hat{u}_2(t) & \dots & \hat{u}_{n_n}(t) \end{bmatrix}^\top = \begin{bmatrix} \hat{u}_D(t) & \hat{u}_\Omega(t) \end{bmatrix}^\top$$

Then:

$$\underbrace{\begin{bmatrix} \hat{M}_{DD} & \hat{M}_{D\Omega} \\ \hat{M}_{\Omega D} & \hat{M}_{\Omega\Omega} \end{bmatrix}}_{\hat{M}} \underbrace{\begin{bmatrix} \ddot{\hat{u}}_D(t) \\ \ddot{\hat{u}}_\Omega(t) \end{bmatrix}}_{\ddot{\hat{u}}(t)} + \underbrace{\begin{bmatrix} \hat{D}_{DD} & \hat{D}_{D\Omega} \\ \hat{D}_{\Omega D} & \hat{D}_{\Omega\Omega} \end{bmatrix}}_{\hat{D}} \underbrace{\begin{bmatrix} \dot{\hat{u}}_D(t) \\ \dot{\hat{u}}_\Omega(t) \end{bmatrix}}_{\dot{\hat{u}}(t)} + \underbrace{\begin{bmatrix} \hat{K}_{DD} & \hat{K}_{D\Omega} \\ \hat{K}_{\Omega D} & \hat{K}_{\Omega\Omega} \end{bmatrix}}_{\hat{K}_r} \underbrace{\begin{bmatrix} \hat{u}_D(t) \\ \hat{u}_\Omega(t) \end{bmatrix}}_{\hat{u}(t)} = \underbrace{\begin{bmatrix} 0 \\ \hat{B}_N \end{bmatrix}}_{\hat{B}} F_L(t)$$

where $\dot{\hat{u}}_D(t) = v_D(t)$ is the Dirichlet B.C. to be imposed.

Since $\hat{p}(t) = \hat{M}\dot{\hat{u}}(t)$ we have:

$$\hat{p}_\Omega(t) = \hat{M}_{\Omega D} v_D(t) + \hat{M}_{\Omega\Omega} \dot{\hat{u}}_\Omega(t) \quad \Longrightarrow \quad \dot{\hat{u}}_\Omega(t) = \hat{M}_{\Omega\Omega}^{-1} \hat{p}_\Omega(t) - \hat{M}_{\Omega\Omega}^{-1} \hat{M}_{\Omega D} v_D(t)$$

Standard finite element method

Substituting the previous expression into the second equation and grouping terms yields:

$$\dot{\hat{p}}_\Omega(t) + \hat{D}_{\Omega\Omega}\hat{M}_{\Omega\Omega}^{-1}\hat{p}_\Omega(t) + \hat{K}_{\Omega D}\hat{u}_D(t) + \hat{K}_{\Omega\Omega}\hat{u}_\Omega(t) = \ldots$$
$$\ldots \hat{B}_N F_L(t) + (\hat{D}_{\Omega\Omega}\hat{M}_{\Omega\Omega}^{-1}\hat{M}_{\Omega D} - \hat{D}_{\Omega D})v_D(t)$$

Standard finite element method

Substituting the previous expression into the second equation and grouping terms yields:

$$\dot{\hat{p}}_\Omega(t) + \hat{D}_{\Omega\Omega}\hat{M}_{\Omega\Omega}^{-1}\hat{p}_\Omega(t) + \hat{K}_{\Omega D}\hat{u}_D(t) + \hat{K}_{\Omega\Omega}\hat{u}_\Omega(t) = \ldots$$
$$\ldots \hat{B}_N F_L(t) + (\hat{D}_{\Omega\Omega}\hat{M}_{\Omega\Omega}^{-1}\hat{M}_{\Omega D} - \hat{D}_{\Omega D})v_D(t)$$

Finally, the resulting PHS model can be written as:

$$\underbrace{\begin{bmatrix}\dot{\hat{p}}_\Omega\\ \dot{\hat{u}}_D\\ \dot{\hat{u}}_\Omega\end{bmatrix}}_{\dot{\hat{x}}} = \underbrace{\begin{bmatrix}-\hat{D}_{\Omega\Omega} & 0 & -I_\Omega\\ 0 & 0 & 0\\ I_\Omega & 0 & 0\end{bmatrix}}_{(\hat{J}-\hat{R})} \underbrace{\begin{bmatrix}\hat{M}_{\Omega\Omega}^{-1}\hat{p}_\Omega\\ \hat{K}_{DD}\hat{u}_D+\hat{K}_{D\Omega}\hat{u}_\Omega\\ \hat{K}_{\Omega D}\hat{u}_D+\hat{K}_{\Omega\Omega}\hat{u}_\Omega\end{bmatrix}}_{\nabla\hat{H}(\hat{x})} + \underbrace{\begin{bmatrix}\hat{B}_N & (\hat{D}_{\Omega\Omega}\hat{M}_{\Omega\Omega}^{-1}\hat{M}_{\Omega D}-\hat{D}_{\Omega D})\\ 0 & I_D\\ 0 & -\hat{M}_{\Omega\Omega}^{-1}\hat{M}_{\Omega D}\end{bmatrix}}_{\hat{G}_\partial} \underbrace{\begin{bmatrix}F_L\\ v_D\end{bmatrix}}_{u_\partial}$$

$$y_\partial = \hat{G}_\partial^\top \nabla\hat{H}(\hat{x})$$

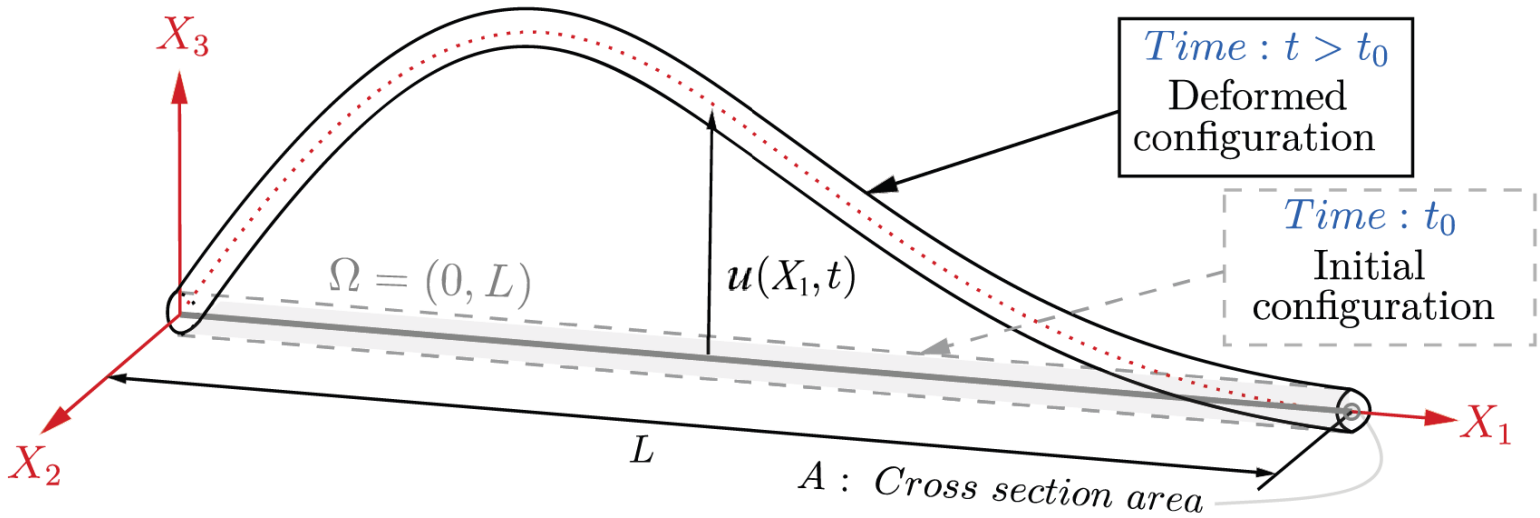
$$\hat{H}(\hat{x}) = \frac{1}{2}\hat{p}_\Omega^\top \hat{M}_{\Omega\Omega}^{-1}\hat{p}_\Omega + \frac{1}{2}\hat{u}^\top \hat{K}_r\hat{u}$$

Example: 1D vibrating string

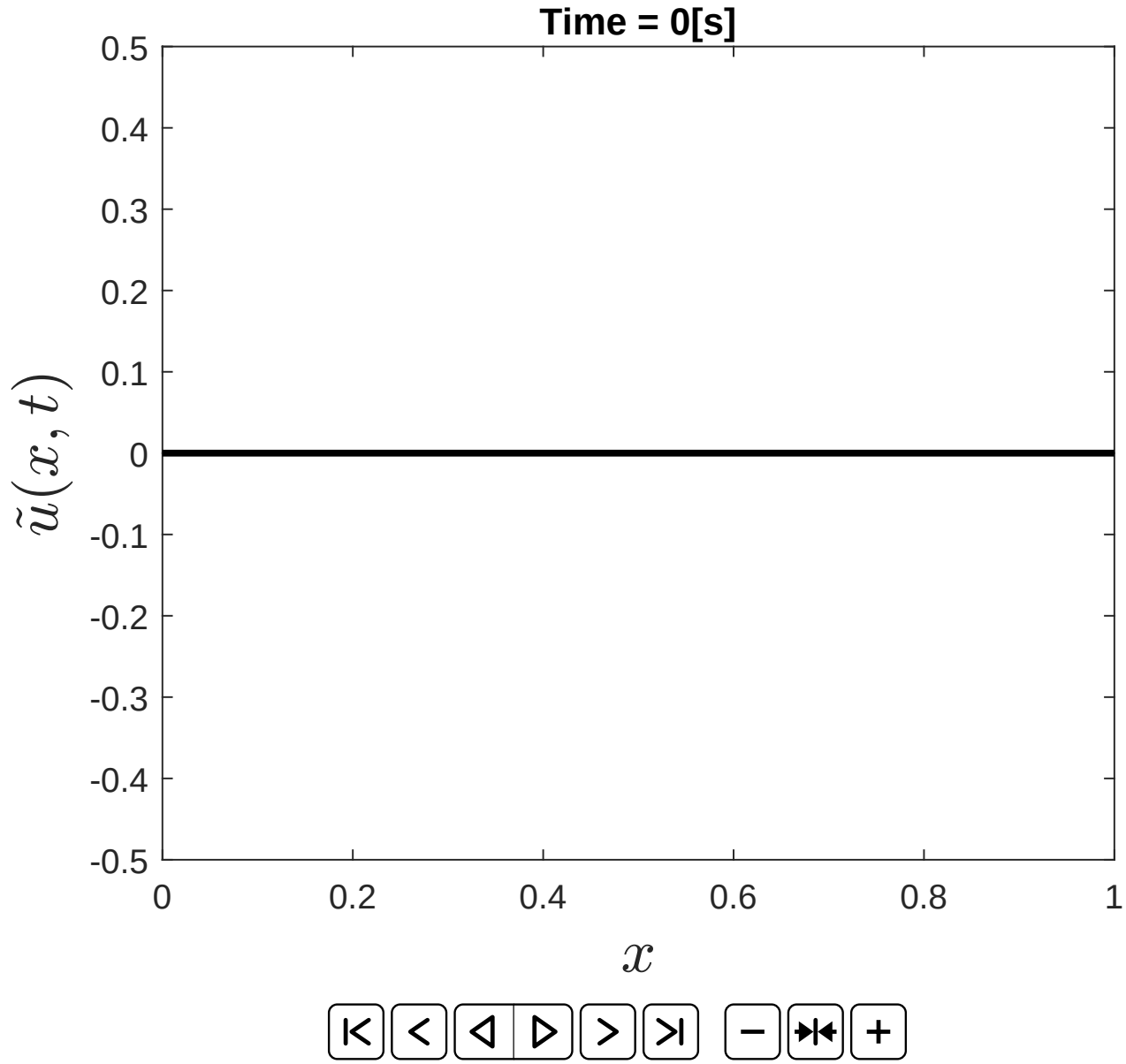
$$\rho A\ddot{u} + d\dot{u} - \frac{\partial}{\partial x}\left(T\frac{\partial u}{\partial x}\right) = 0 \quad \text{(PDE)}$$

$$\dot{u}(x,t)\Big|_{x=0} = 0 \quad \text{(Dirichlet B.C.)}$$

$$T\frac{\partial u}{\partial x}\Big|_{x=L} = F_L(t) \quad \text{(Neumann B.C.)}$$



Example: 1D vibrating string



Mixed finite element method

Hu-Washizu principle:

- $$\delta P_r = \int_\Omega \delta \dot{r}^\top \left(\mathcal{M}\ddot{r} + \mathcal{F}^* e_\epsilon\right) d\mathtt{X} + \int_{\partial\Omega_N} \delta \dot{r}^\top \left(F_\partial e_\epsilon - \tau_N\right) d\mathtt{S} = 0$$
- $$\delta P_{e_\epsilon} = \int_\Omega \delta e_\epsilon^\top \left(\dot{\epsilon} - \mathcal{F}\dot{r}\right) d\mathtt{X} = 0$$
- $$\delta P_\epsilon = \int_\Omega \delta \dot{\epsilon}^\top \left(e_\epsilon - \mathcal{K}_\epsilon \epsilon\right) d\mathtt{X} = 0$$

with independent variations of $(\delta\dot{r}, \delta\epsilon_\epsilon, \delta\epsilon)$, and subject to:

$$\delta\dot{r}(\mathtt{S}, t) = 0, \quad \forall \mathtt{S} \in \partial\Omega_D.$$

Mixed finite element method

Hu-Washizu principle:

- $\delta P_r = \int_\Omega \delta \dot{r}^\top \left(\mathcal{M}\ddot{r} + \mathcal{F}^* e_\epsilon\right) d\mathtt{X} + \int_{\partial\Omega_N} \delta \dot{r}^\top \left(F_\partial e_\epsilon - \tau_N\right) d\mathtt{S} = 0$
- $\delta P_{e_\epsilon} = \int_\Omega \delta e_\epsilon^\top \left(\dot{\epsilon} - \mathcal{F}\dot{r}\right) d\mathtt{X} = 0$
- $\delta P_\epsilon = \int_\Omega \delta \dot{\epsilon}^\top \left(e_\epsilon - \mathcal{K}_\epsilon \epsilon\right) d\mathtt{X} = 0$

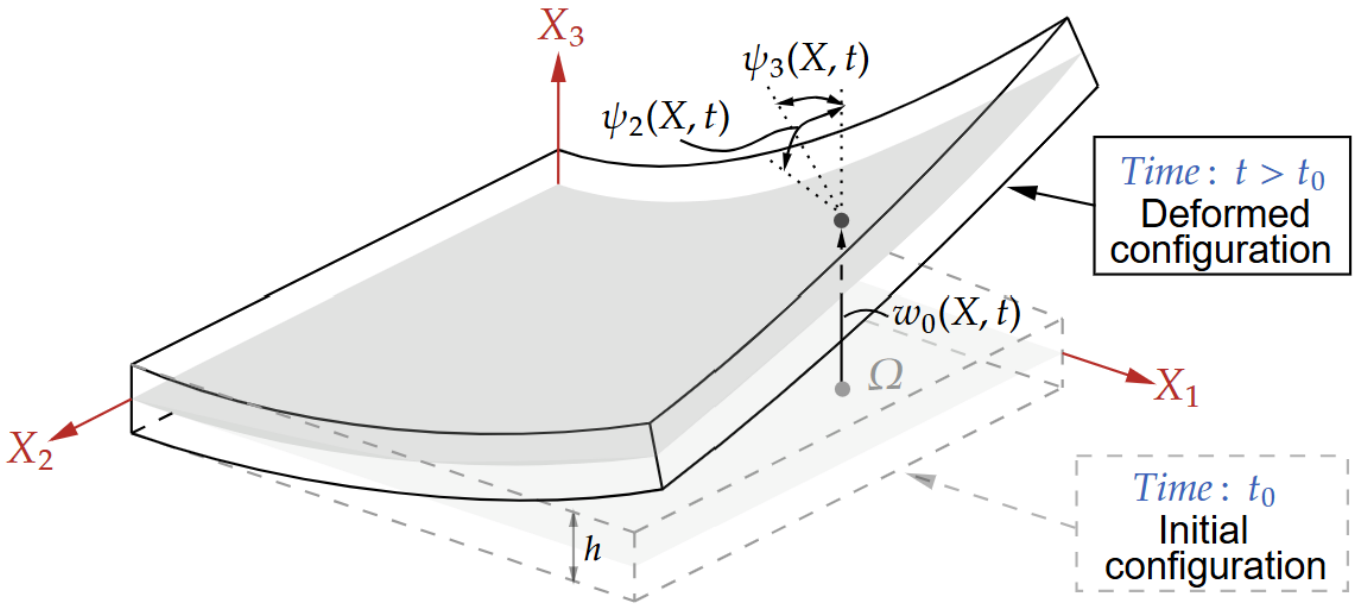
with independent variations of $(\delta\dot{r}, \delta\epsilon_\epsilon, \delta\epsilon)$, and subject to:

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Remarks:

1) This method preserves the PHS structure (2) and prevents shear locking, making it ideal for bending-dominated problems (such as beams and plates).
2) May introduce non-physical vibration modes.

Example: Mindlin plate



Displacement field:

$$\mathbf{u}(\mathbf{X},t) = \underbrace{\begin{bmatrix} 0 & -X_3 & 0 \\ -X_3 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\bar{M}_1(\mathtt{x}^c)} \underbrace{\begin{bmatrix} \psi_2(\mathtt{x},t) \\ \psi_3(\mathtt{x},t) \\ w_0(\mathtt{x},t) \end{bmatrix}}_{r(\mathtt{x},t)} \quad , \quad \mathtt{x} = (X_1, X_2)$$

Example: Mindlin plate

$$\frac{\rho h^3}{12}\ddot{\psi} - \mathrm{Div}(\underline{\underline{D}} : \mathrm{Grad}(\psi)) - \kappa G h\,(\mathrm{grad}(w) - \psi) = 0 \qquad \text{(PDE 1)}$$

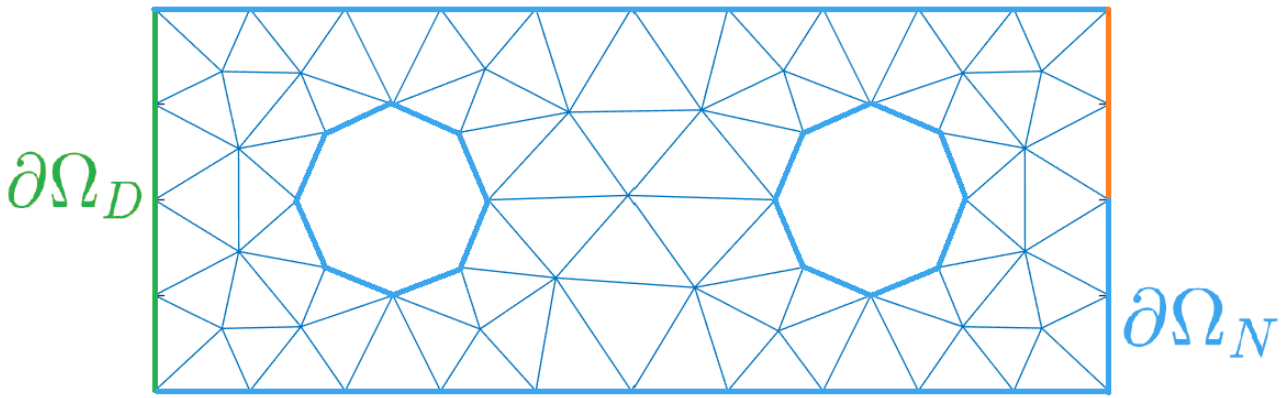
$$\rho h \ddot{w} - \mathrm{div}(\kappa G h\,(\mathrm{grad}(w) - \psi)) = 0 \qquad \text{(PDE 2)}$$

$$\dot{\psi}(x, y, t)\Big|_{\partial\Omega_D} = 0 \qquad \text{(Dirichlet B.C. 1)}$$

$$\dot{w}(x, y, t)\Big|_{\partial\Omega_D} = 0 \qquad \text{(Dirichlet B.C. 2)}$$

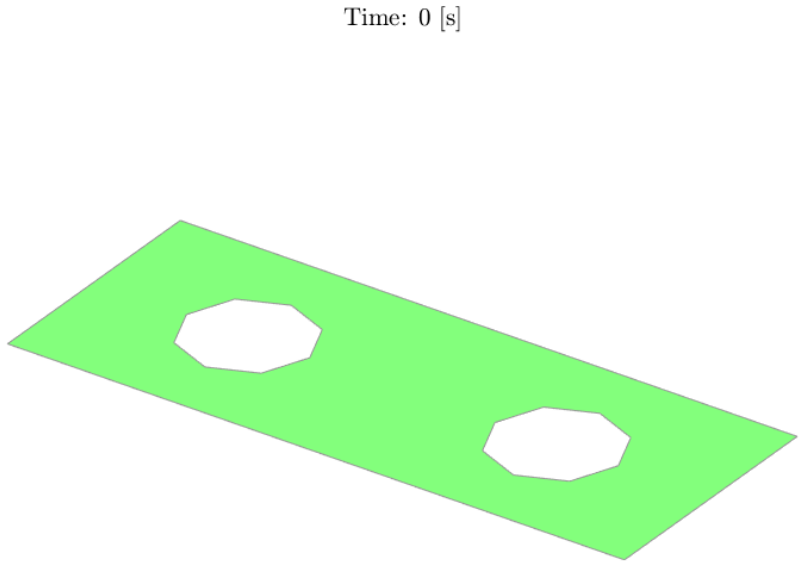
$$\underline{\underline{D}} : \mathrm{Grad}(\psi)\hat{n}\Big|_{\partial\Omega_N} = 0 \qquad \text{(Neumann B.C. 1)}$$

$$\kappa G h\,(\mathrm{grad}(w) - \psi) \cdot \hat{n}\Big|_{\partial\Omega_N} = F_N(S, t) \qquad \text{(Neumann B.C. 2)}$$



Example: Mindlin plate

mFEM
C – C



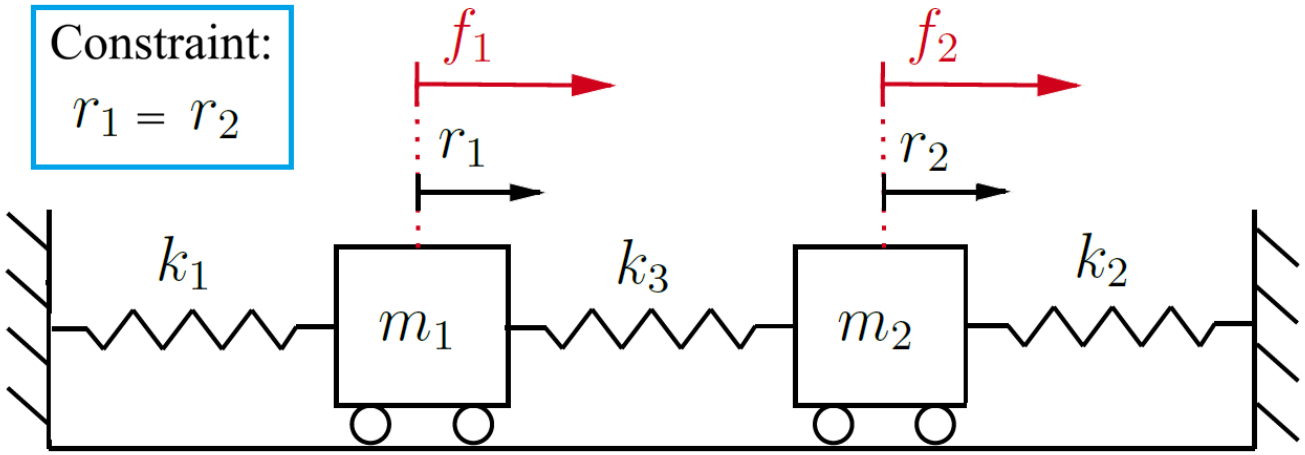
Thank you for your attention.

Any questions?

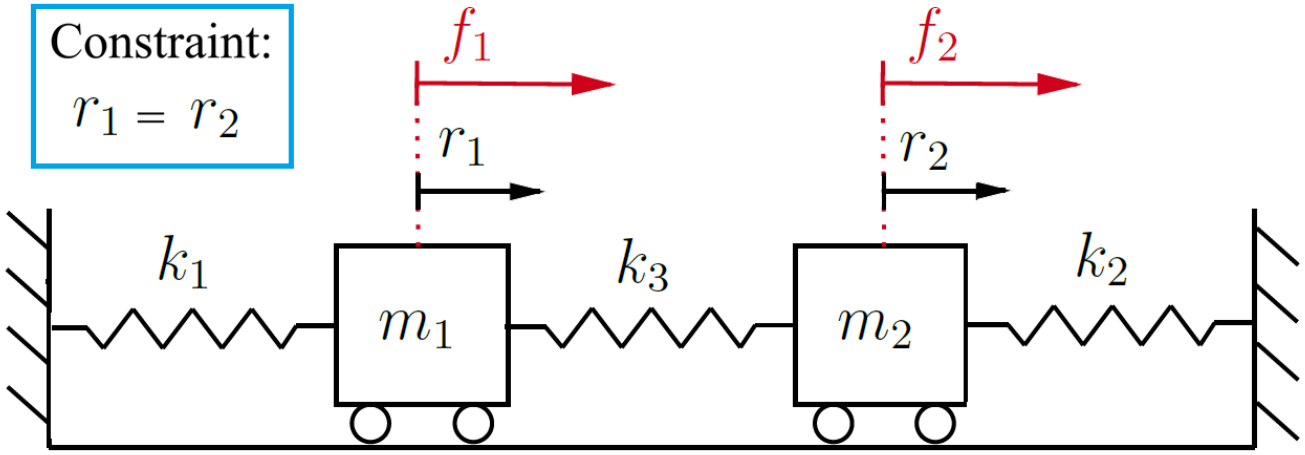
cristobal.ponces@usm.cl

Additional slides

Lumped parameter system with constraint

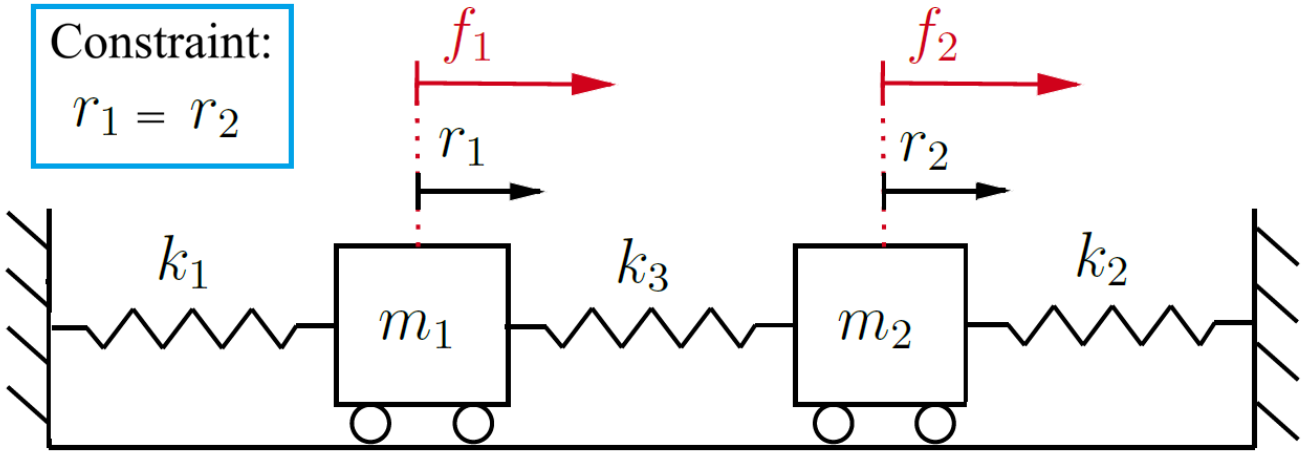


Lumped parameter system with constraint



$$T = \frac{1}{2}\begin{bmatrix}\dot{r}_1\\ \dot{r}_2\end{bmatrix}^\top \underbrace{\begin{bmatrix}m_1 & 0\\ 0 & m_2\end{bmatrix}}_{M}\begin{bmatrix}\dot{r}_1\\ \dot{r}_2\end{bmatrix},\quad U = \frac{1}{2}\begin{bmatrix}r_1\\ r_2\end{bmatrix}^\top \underbrace{\begin{bmatrix}k_1+k_3 & -k_3\\ -k_3 & k_2+k_3\end{bmatrix}}_{K}\begin{bmatrix}r_1\\ r_2\end{bmatrix},\quad W_E = \begin{bmatrix}r_1\\ r_2\end{bmatrix}^\top \underbrace{\begin{bmatrix}f_1\\ f_2\end{bmatrix}}_{F}$$

Lumped parameter system with constraint

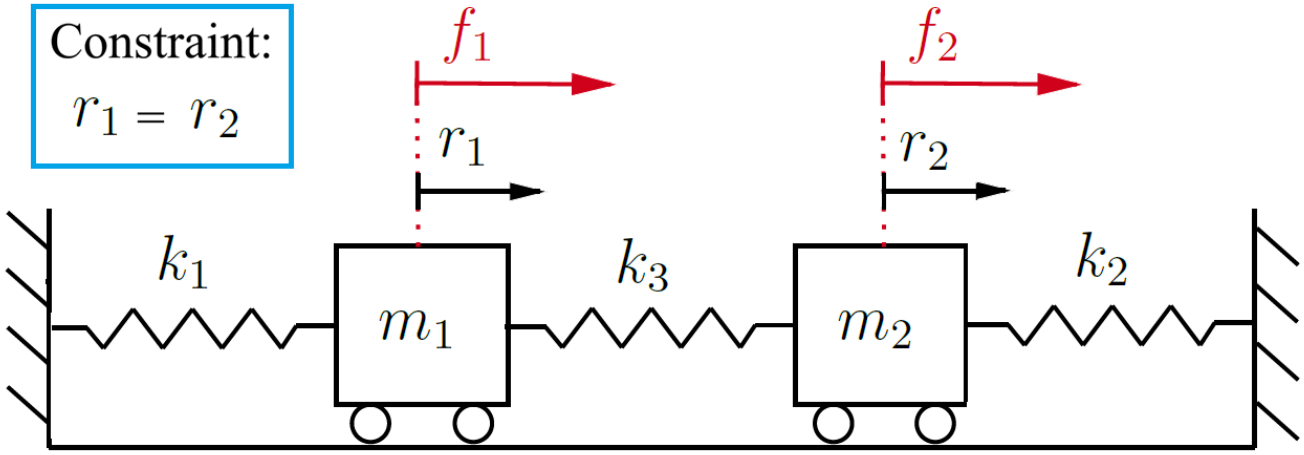


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Constraint:

$$\gamma(r) = r_1 - r_2 = 0$$

Lumped parameter system with constraint



$$T = \frac{1}{2}\begin{bmatrix}\dot{r}_1\\ \dot{r}_2\end{bmatrix}^\top \underbrace{\begin{bmatrix}m_1 & 0\\ 0 & m_2\end{bmatrix}}_{M}\begin{bmatrix}\dot{r}_1\\ \dot{r}_2\end{bmatrix}, \quad U = \frac{1}{2}\begin{bmatrix}r_1\\ r_2\end{bmatrix}^\top \underbrace{\begin{bmatrix}k_1+k_3 & -k_3\\ -k_3 & k_2+k_3\end{bmatrix}}_{K}\begin{bmatrix}r_1\\ r_2\end{bmatrix}, \quad W_E = \begin{bmatrix}r_1\\ r_2\end{bmatrix}^\top \underbrace{\begin{bmatrix}f_1\\ f_2\end{bmatrix}}_{F}$$

Constraint:

$$\gamma(r) = r_1 - r_2 = 0 \quad \Longleftrightarrow \quad \gamma(r) = \underbrace{\begin{bmatrix}1 & -1\end{bmatrix}}_{G_\lambda}\begin{bmatrix}r_1\\ r_2\end{bmatrix} = 0$$

Extended Hamilton's principle

$$\int_{t_1}^{t_2} [\delta(T - U) + \delta W_E - \delta C]\, dt = 0, \tag{10}$$

$$\delta r(t_1) = \delta r(t_2) = 0, \tag{11}$$

where $C = \lambda^\top \gamma(r)$.

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For the previous system we have:

$$\delta T = \delta \dot{r}^\top M \dot{r}, \quad \delta U = \delta r^\top K r, \quad \delta W_E = \delta r^\top F, \quad \delta C = \delta \lambda^\top G_\lambda r + \delta r^\top G_\lambda^\top \lambda.$$

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Replacing above:

$$\rightarrow \int_{t_1}^{t_2} [\delta \dot{r}^\top M \dot{r} - \delta r^\top K r + \delta r^\top F - \delta \lambda^\top G_\lambda r - \delta r^\top G_\lambda^\top \lambda]\, dt = 0$$

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$$\rightarrow \int_{t_1}^{t_2} [\delta r^\top (-M\ddot{r} - Kr - G_\lambda^\top \lambda + F) + \delta\lambda^\top G_\lambda r]\, dt + (\delta r^\top M \dot{r})\Big|_{t_1}^{t_2} = 0$$

Extended Hamilton's principle

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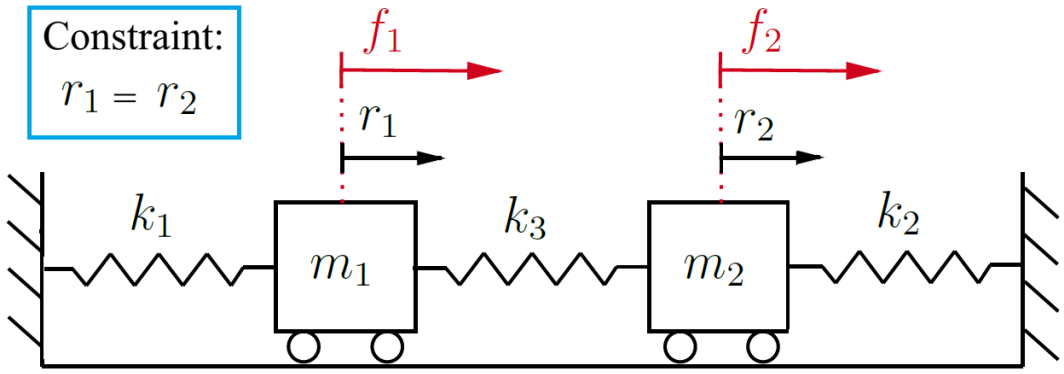
$$\rightarrow \int_{t_1}^{t_2} [\delta r^\top (-M\ddot{r} - Kr - G_\lambda^\top \lambda + F) + \delta\lambda^\top G_\lambda r]\, dt + (\delta r^\top M \dot{r})\Big|_{t_1}^{t_2} = 0$$

Since δr and $\delta\lambda$ are arbitrary, then we have:

$$M\ddot{r} + Kr + G_\lambda^\top \lambda = F \tag{12}$$

$$G_\lambda r = 0 \tag{13}$$

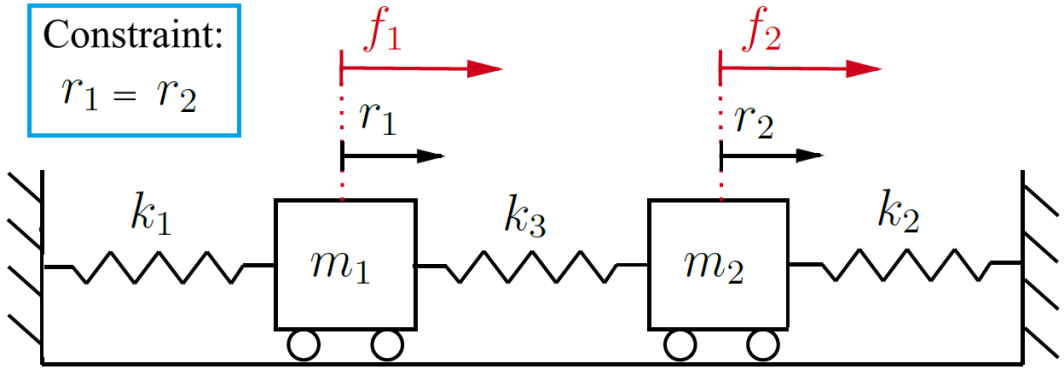
Lumped parameter system with constraint: PH-DAE representation



$$M\ddot{r} + Kr + G_\lambda^\top \lambda = F \tag{12}$$

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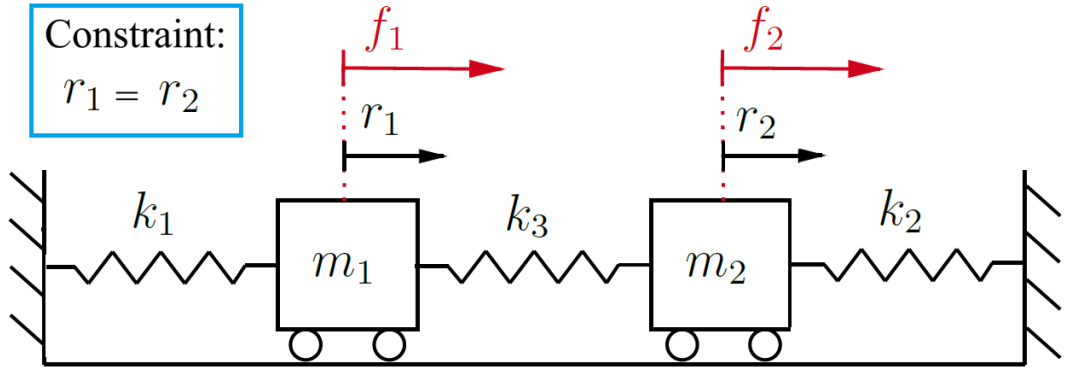
Lumped parameter system with constraint: PH-DAE representation



$$M\ddot{r} + Kr + G_\lambda^\top \lambda = F \tag{12}$$

$$G_\lambda r = 0 \quad \xrightarrow{\partial/\partial t} \quad G_\lambda \dot{r} = 0 \tag{13}$$

Lumped parameter system with constraint: PH-DAE representation



$$M\ddot{r} + Kr + G_\lambda^\top \lambda = F \tag{12}$$

$$G_\lambda r = 0 \quad \xrightarrow{\partial/\partial t} \quad G_\lambda \dot{r} = 0 \tag{13}$$

Defining $p = M\dot{r}$, the previous model can be rewritten as:

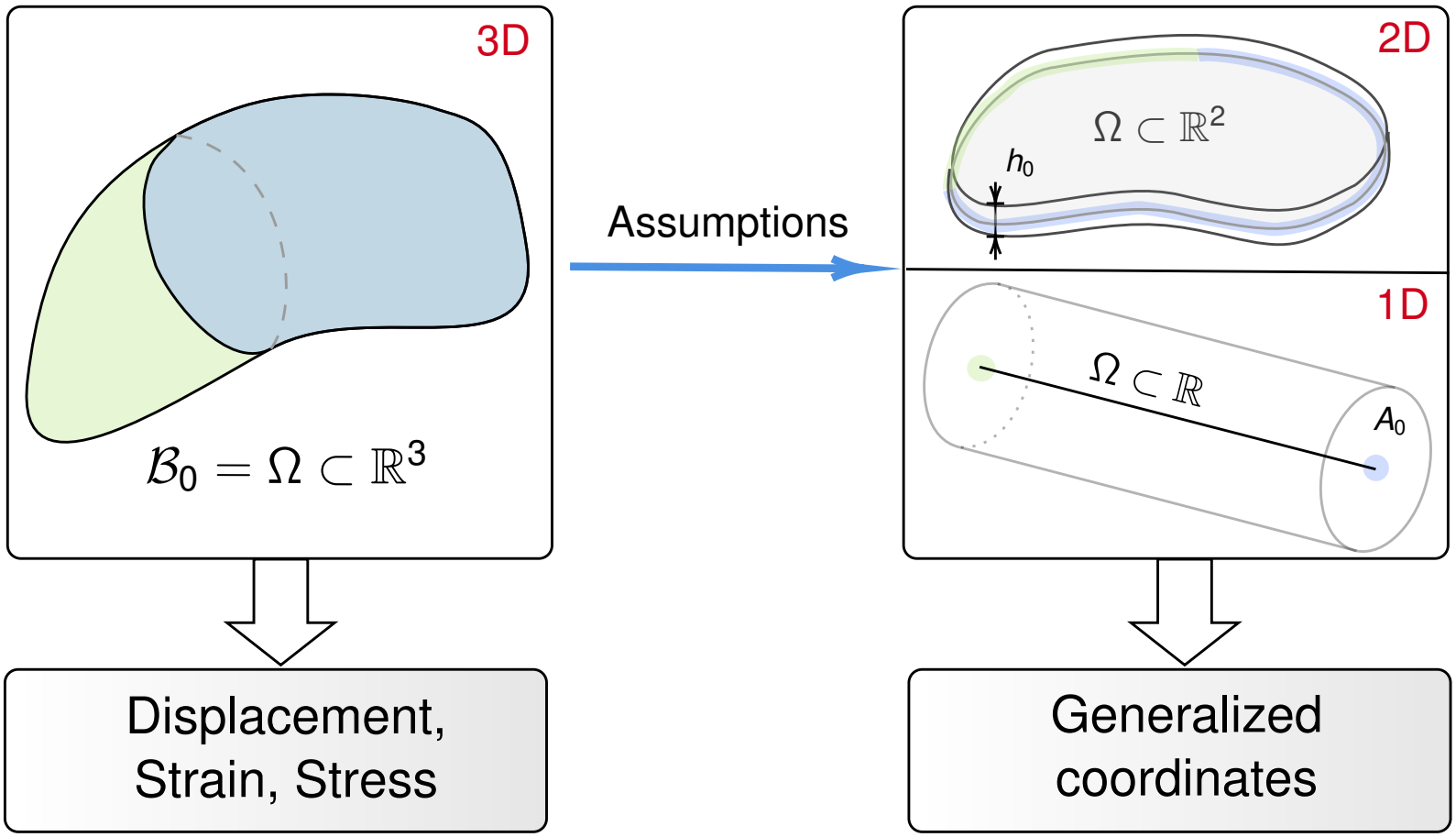
$$\underbrace{\begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{E} \underbrace{\begin{bmatrix} \dot{p} \\ \dot{r} \\ \dot{\lambda} \end{bmatrix}}_{\dot{x}} = \underbrace{\begin{bmatrix} 0 & -I & -G_\lambda^\top \\ I & 0 & 0 \\ G_\lambda & 0 & 0 \end{bmatrix}}_{J=-J^\top} \underbrace{\begin{bmatrix} M^{-1}p \\ Kr \\ \lambda \end{bmatrix}}_{Z(x)} + \underbrace{\begin{bmatrix} I \\ 0 \\ 0 \end{bmatrix}}_{G} F$$

$$y = G^\top \nabla H(x)$$

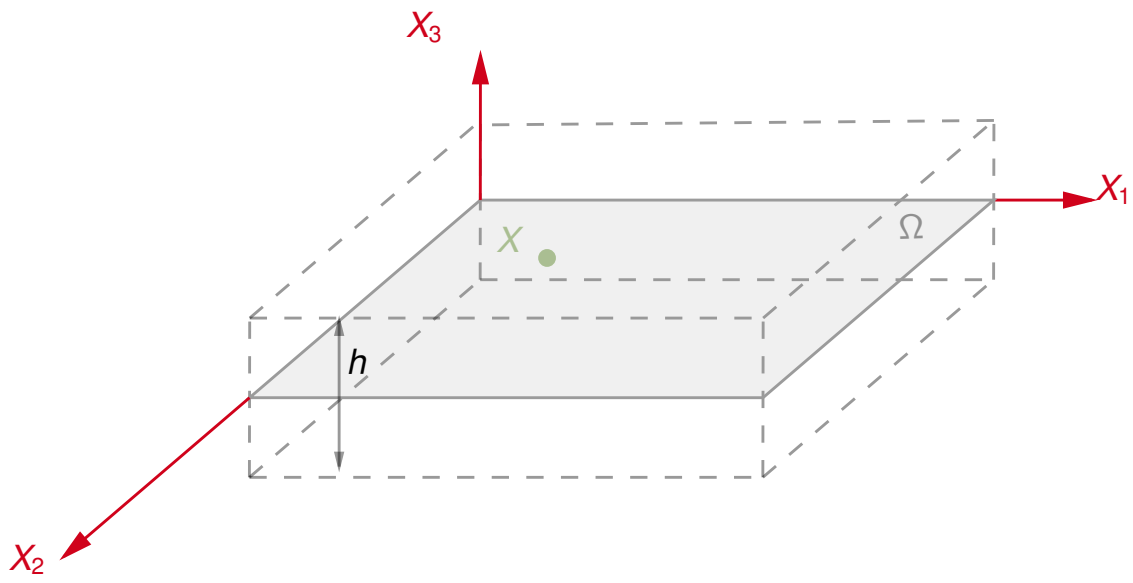
$$H(x) = \tfrac{1}{2} p^\top M^{-1} p + \tfrac{1}{2} r^\top K r$$

with $\nabla H(x) = E^\top Z(x)$.

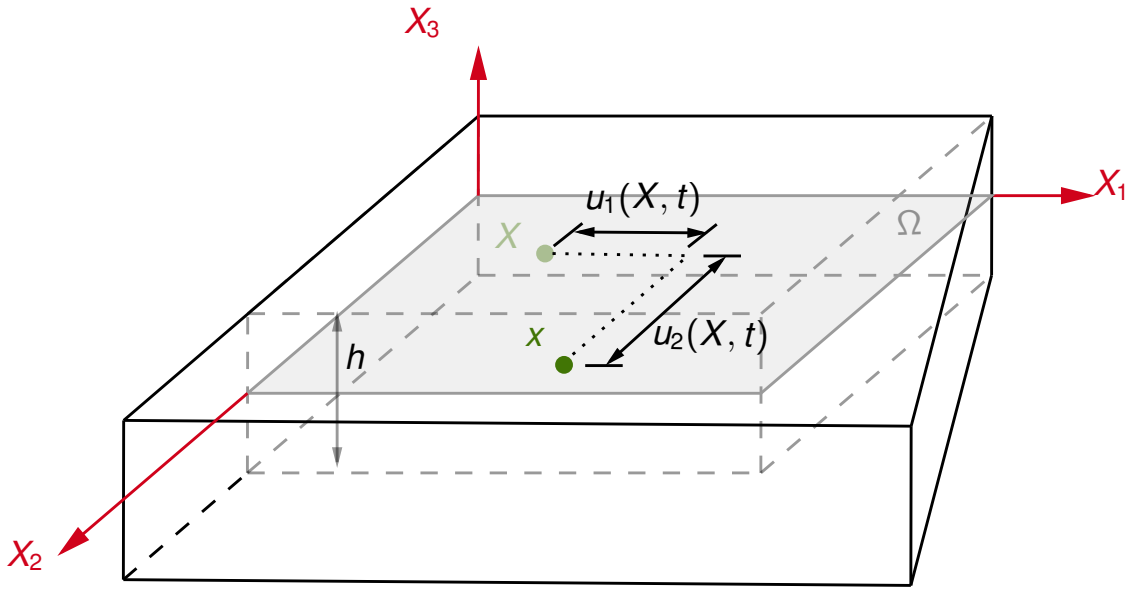
Modeling: General idea



Example: 2D elasticity



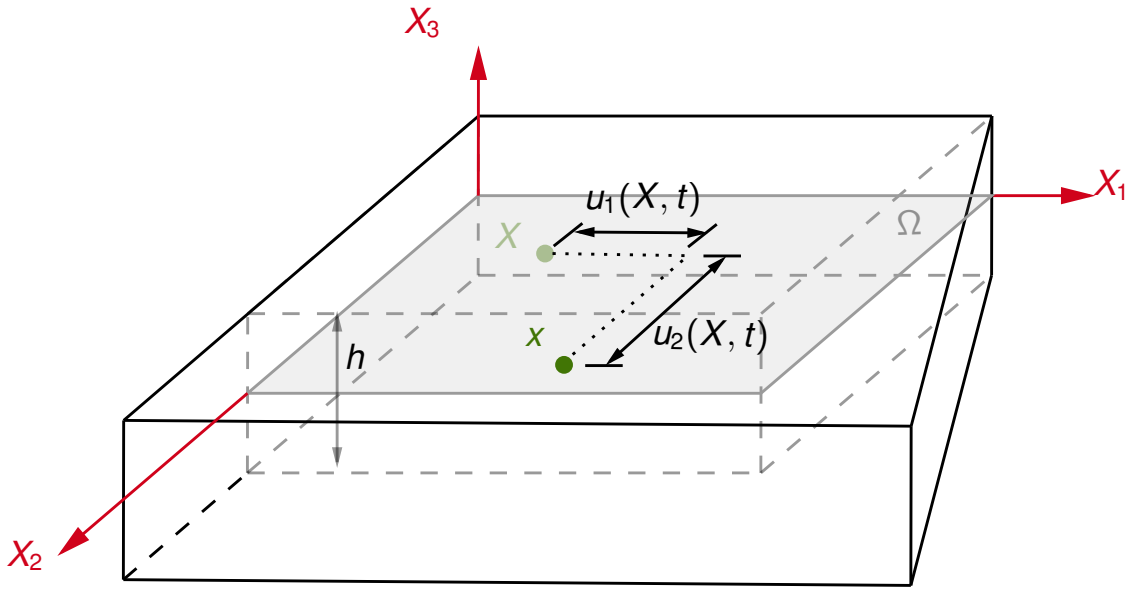
Example: 2D elasticity



Displacement field:

$$\mathbf{u}(\mathbf{X},t) = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}}_{\bar{M}_1(\mathtt{x}^c)} \underbrace{\begin{bmatrix} u_1(\mathtt{X},t) \\ u_2(\mathtt{X},t) \end{bmatrix}}_{r(\mathtt{X},t)}$$

Example: 2D elasticity

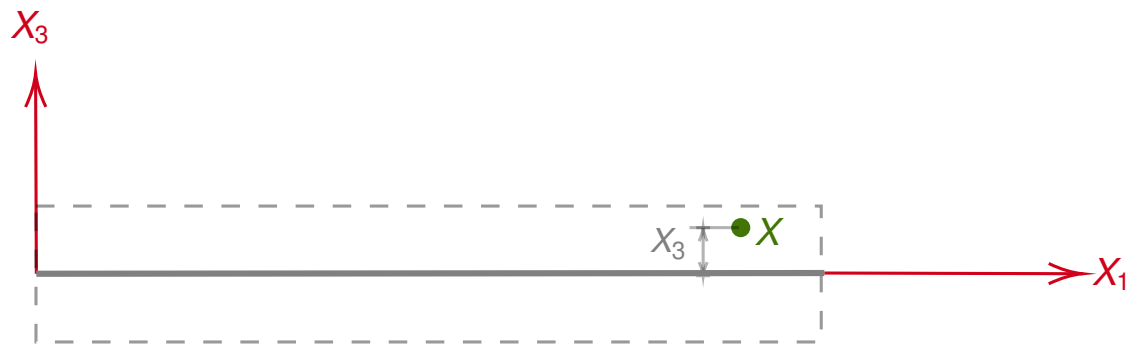


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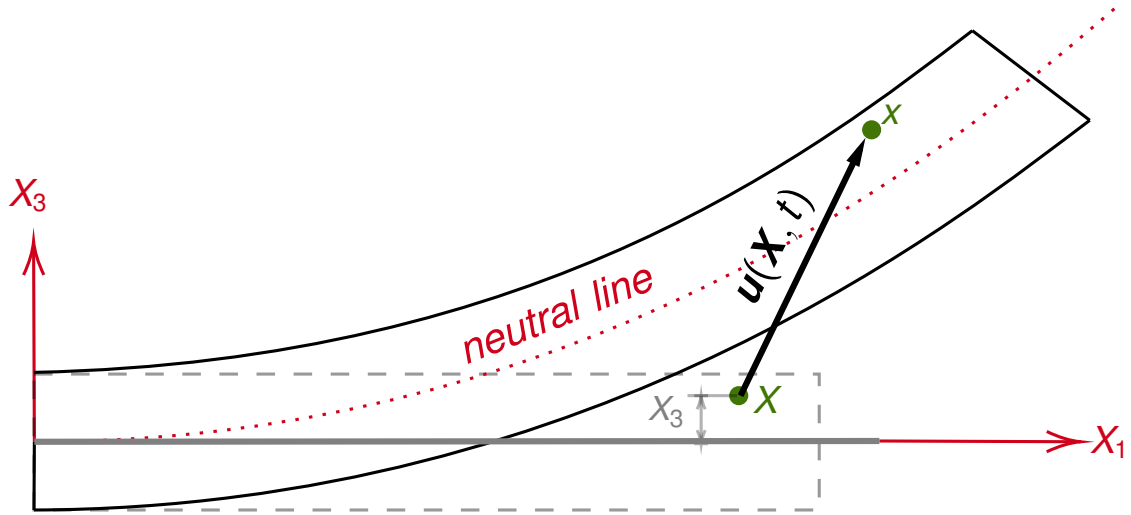
Example: Planar Timoshenko beam

Kinematic assumption: Plane sections normal to the neutral line before deformation remain plane but not necessarily normal to the neutral line after deformation.



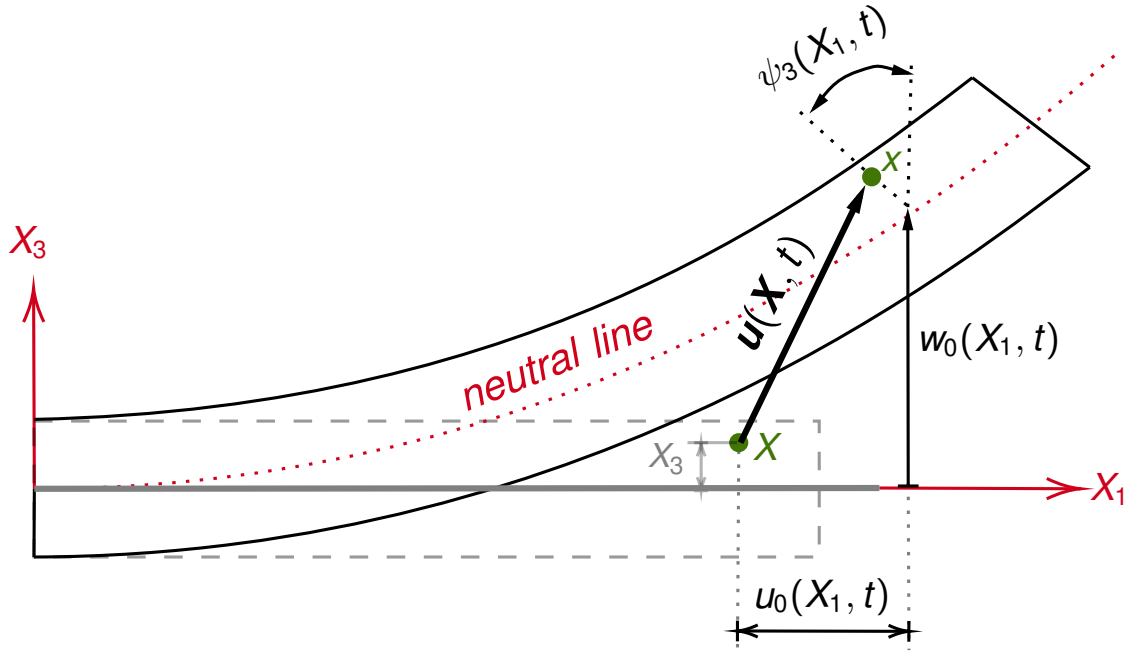
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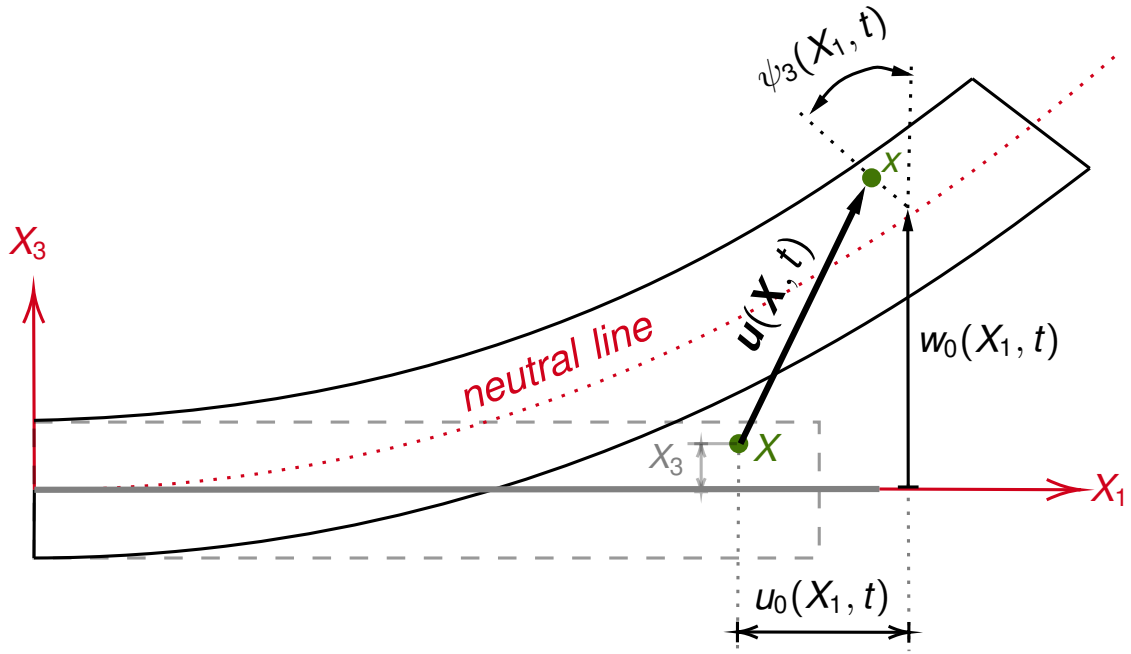
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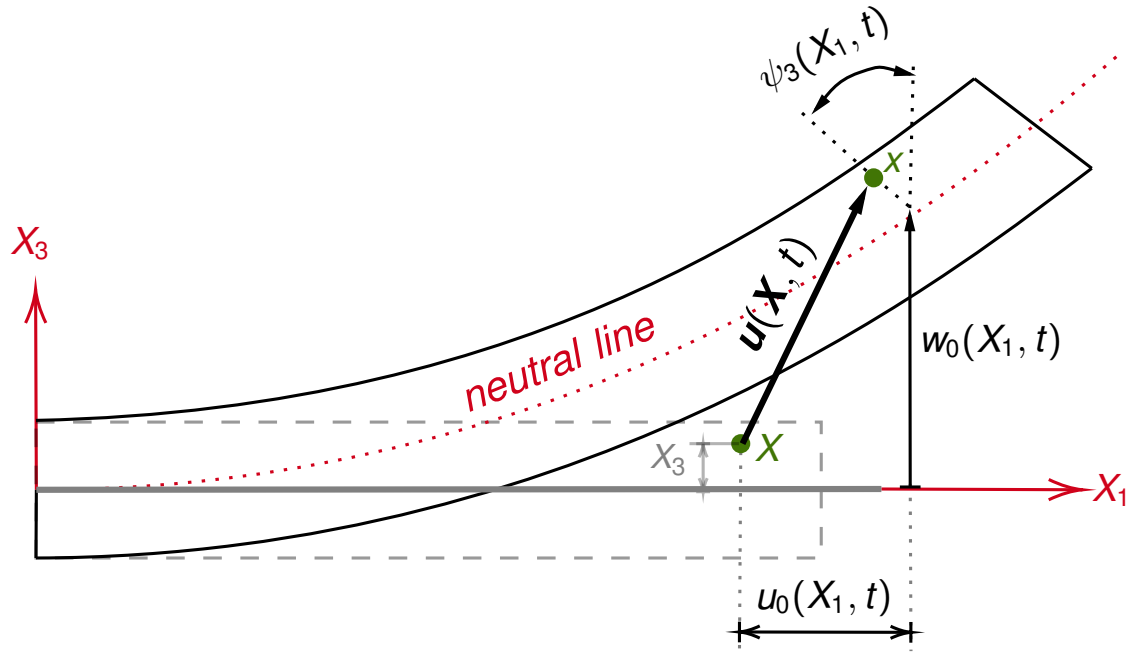
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$$\boldsymbol{u}(\boldsymbol{X}, t) = \underbrace{\begin{bmatrix} 1 & -X_3 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\overline{M}_1(\mathrm{x}^c)} \underbrace{\begin{bmatrix} u_0(\mathrm{x}, t) \\ \psi_3(\mathrm{x}, t) \\ w_0(\mathrm{x}, t) \end{bmatrix}}_{r(\mathrm{x},t)}$$

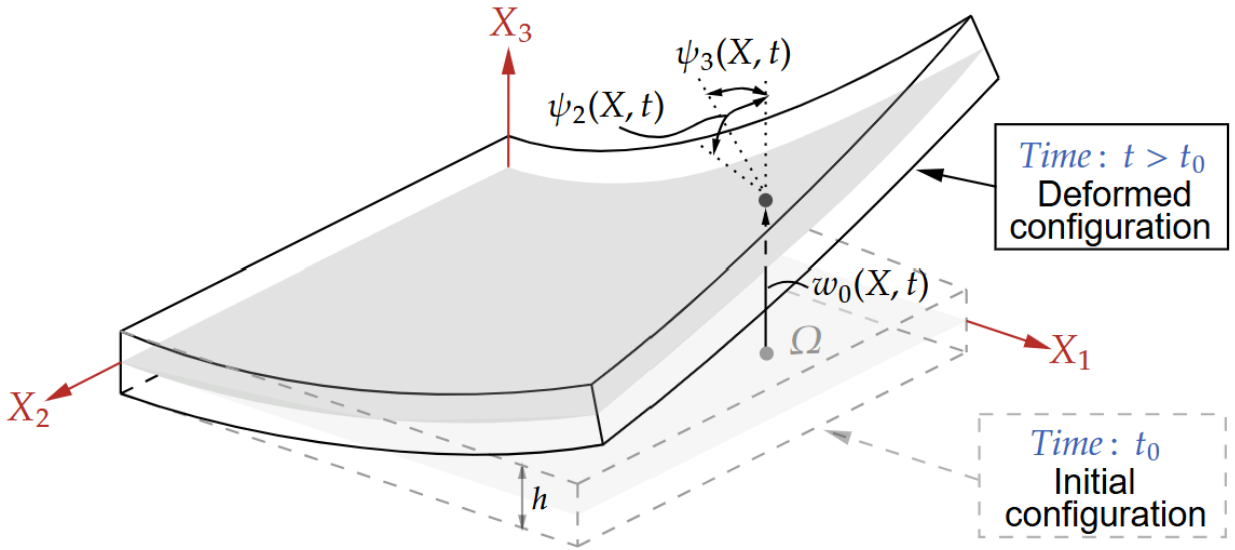
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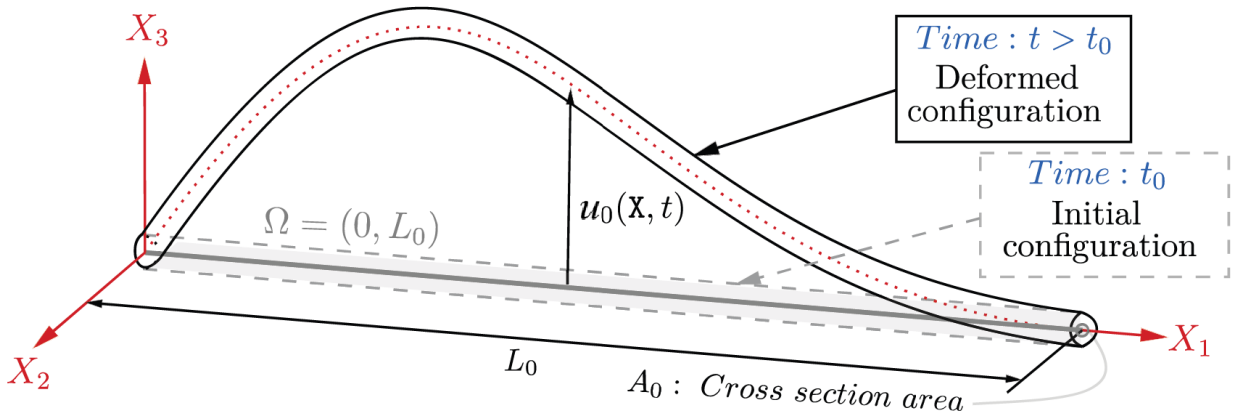
Example: Mindlin plate



Displacement field:

$$\mathbf{u}(\mathbf{X},t) = \underbrace{\begin{bmatrix} 0 & -X_3 & 0 \\ -X_3 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\bar{M}_1(\mathtt{x}^c)} \underbrace{\begin{bmatrix} \psi_2(\mathtt{x},t) \\ \psi_3(\mathtt{x},t) \\ w_0(\mathtt{x},t) \end{bmatrix}}_{r(\mathtt{x},t)} \quad , \quad \mathtt{x} = (X_1, X_2)$$

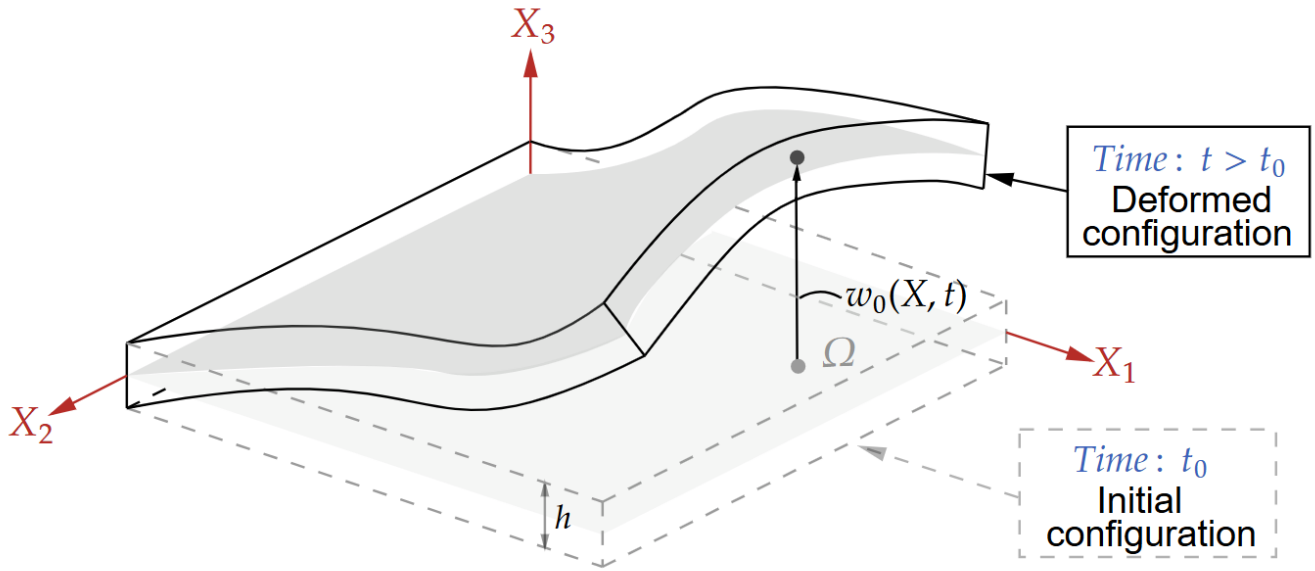
Example: 1D vibrating string



Displacement field:

$$\mathbf{u}(\mathbf{X}, t) = \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}_{\bar{M}_1(\mathtt{X}^c)} \underbrace{\left[u_0(\mathtt{X}, t)\right]}_{r(\mathtt{X},t)} \quad , \quad \mathtt{X} = X_1$$

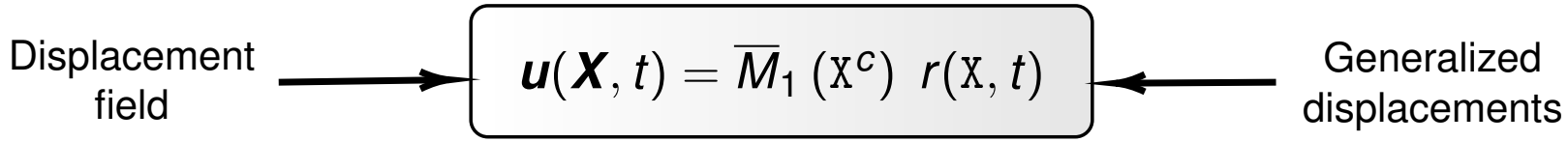
Example: 2D vibrating membrane



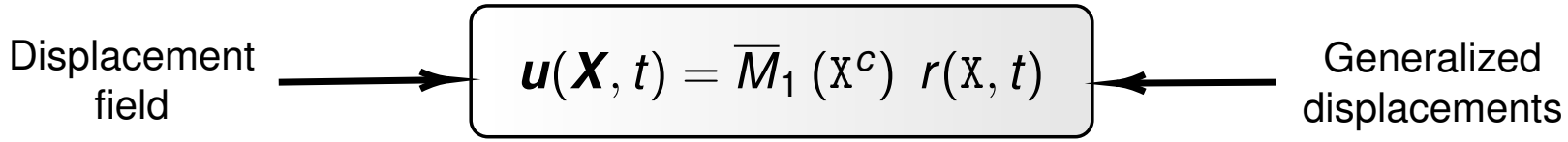
Displacement field:

$$\mathbf{u}(\mathbf{X},t) = \underbrace{\begin{bmatrix}0\\0\\1\end{bmatrix}}_{\bar{M}_1(\mathtt{x}^c)} \underbrace{\left[w_0(\mathtt{x},t)\right]}_{r(\mathtt{x},t)} \quad , \quad \mathtt{x} = (X_1, X_2)$$

Modeling: General idea



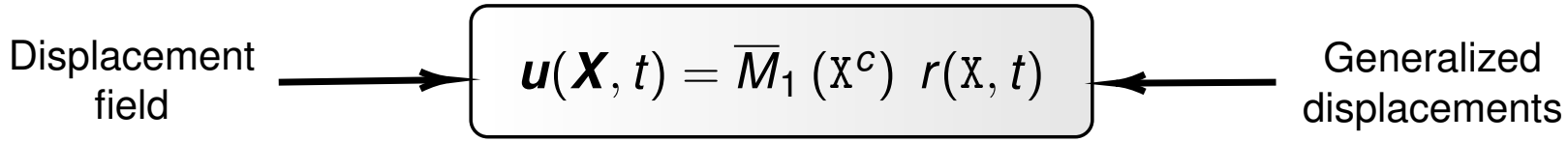
Modeling: General idea



- **Kinetic energy:**

$$T = \frac{1}{2} \int_{\mathcal{B}_0} \rho\, \dot{u}^\top \dot{u}\, dX$$

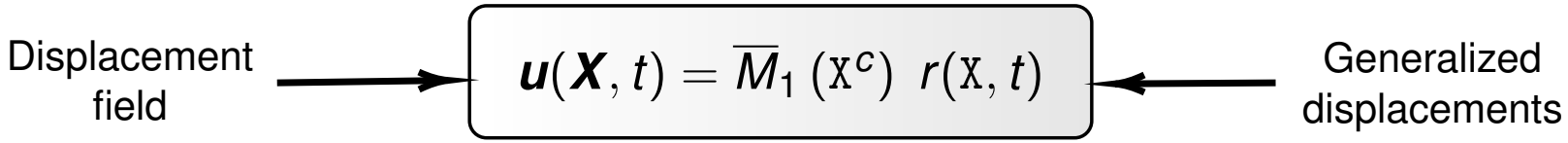
Modeling: General idea



- **Kinetic energy:**

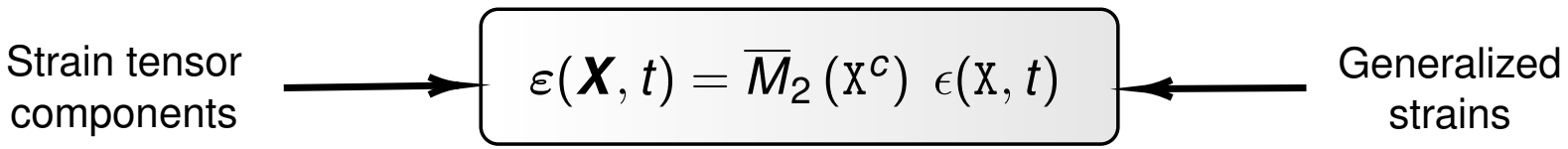
$$T = \frac{1}{2}\int_{\mathcal{B}_0} \rho\, \dot{u}^\top \dot{u}\, dX = \frac{1}{2}\int_{\Omega} \dot{r}^\top \underbrace{\rho \int_{\Omega^c} \bar{M}_1^\top \bar{M}_1\, d\mathrm{X}^c}_{\mathcal{M}(\mathrm{X})} \dot{r}\, d\mathrm{X}$$

Modeling: General idea

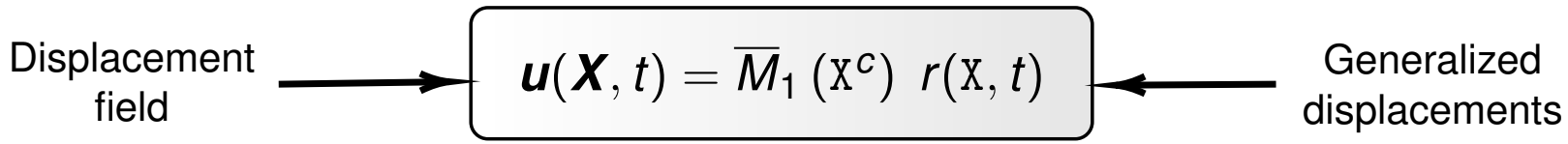


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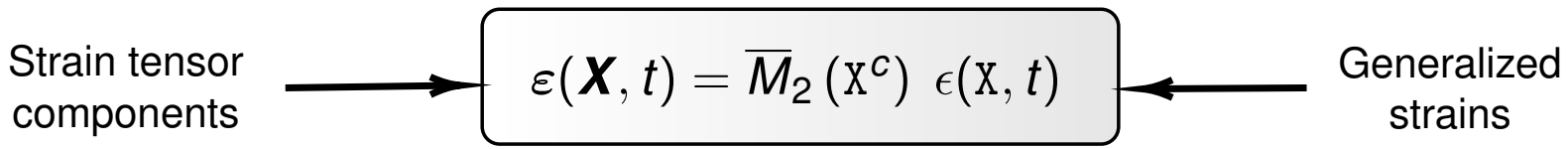


Modeling: General idea



- **Kinetic energy:**

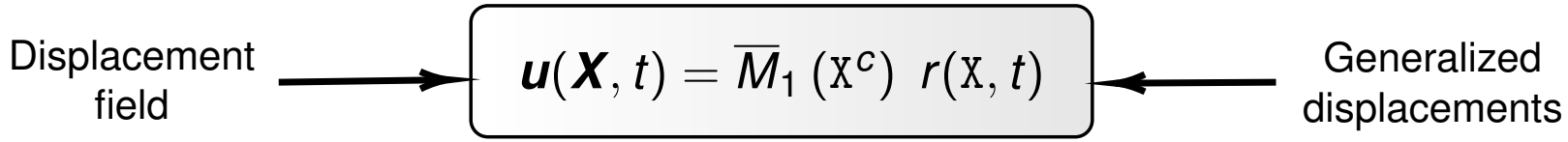
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- **Kinematic equation:**

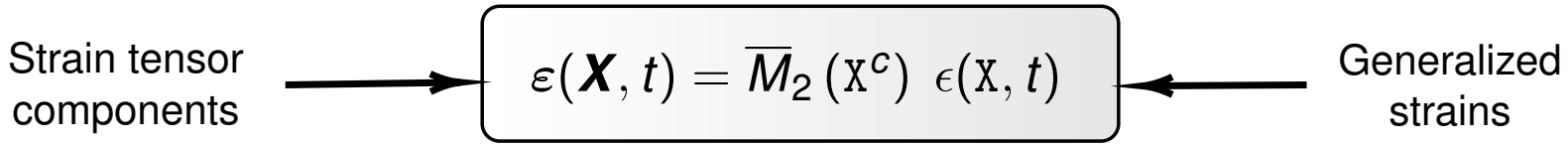
$$\underline{\varepsilon}(X,t) = \frac{1}{2}(\nabla u + \nabla u^\top)$$

Modeling: General idea



- **Kinetic energy:**

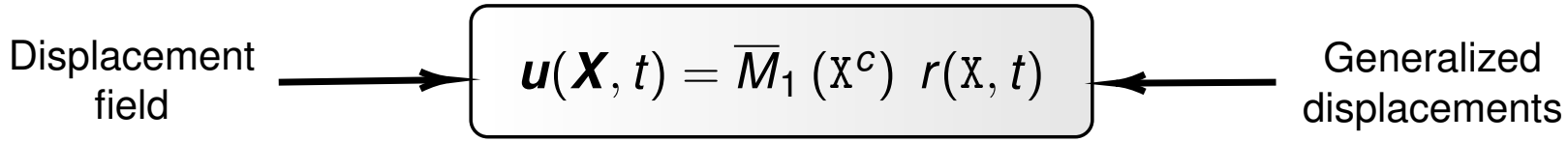
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- **Kinematic equation:**

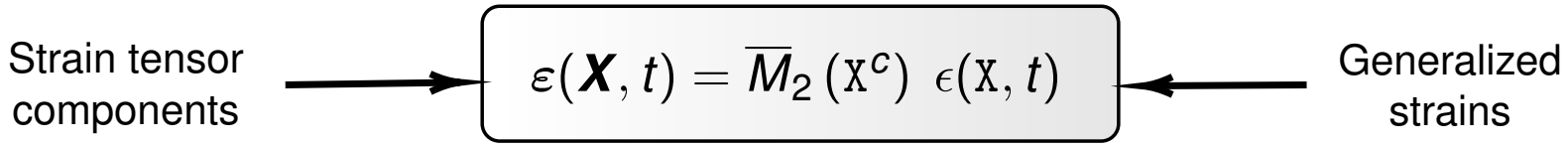
$$\underline{\varepsilon}(X,t) = \frac{1}{2}(\nabla u + \nabla u^\top) \quad \rightarrow \quad \varepsilon(X,t) = \bar{M}_2(\mathtt{X}^c)\mathcal{F}r(\mathtt{X},t)$$

Modeling: General idea



- **Kinetic energy:**

$$T = \frac{1}{2}\int_{\mathcal{B}_0} \rho\, \dot{u}^\top \dot{u}\, dX = \frac{1}{2}\int_{\Omega} \dot{r}^\top \underbrace{\rho \int_{\Omega^c} \bar{M}_1^\top \bar{M}_1\, d\mathtt{X}^c}_{\mathcal{M}(\mathtt{X})}\, \dot{r}\, d\mathtt{X}$$



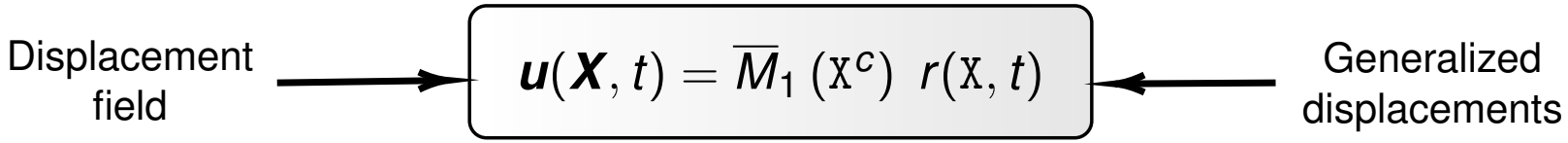
- **Kinematic equation:**

$$\underline{\varepsilon}(X,t) = \frac{1}{2}(\nabla u + \nabla u^\top) \quad \rightarrow \quad \varepsilon(X,t) = \bar{M}_2(\mathtt{X}^c)\mathcal{F} r(\mathtt{X},t)$$

- **Elastic energy:**

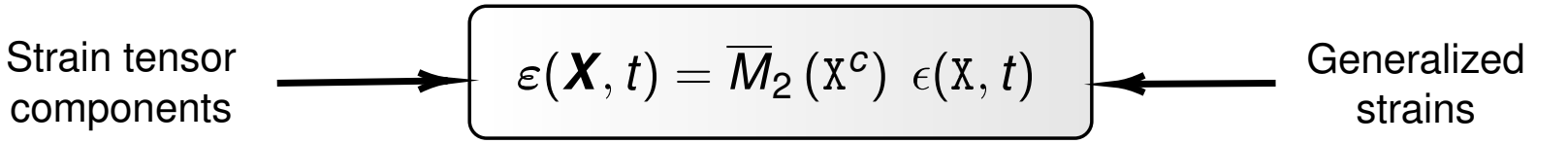
$$U = \frac{1}{2}\int_{\mathcal{B}_0} \varepsilon^\top C\, \varepsilon\, dX$$

Modeling: General idea



- **Kinetic energy:**

$$T = \frac{1}{2}\int_{\mathcal{B}_0} \rho\, \dot{u}^\top \dot{u}\, dX = \frac{1}{2}\int_{\Omega} \dot{r}^\top \underbrace{\rho \int_{\Omega^c} \bar{M}_1^\top \bar{M}_1\, d\mathtt{X}^c}_{\mathcal{M}(\mathtt{X})}\, \dot{r}\, d\mathtt{X}$$



- **Kinematic equation:**

$$\underline{\varepsilon}(X,t) = \frac{1}{2}(\nabla u + \nabla u^\top) \quad \rightarrow \quad \varepsilon(X,t) = \bar{M}_2(\mathtt{X}^c)\mathcal{F} r(\mathtt{X},t)$$

- **Elastic energy:**

$$U = \frac{1}{2}\int_{\mathcal{B}_0} \varepsilon^\top C\, \varepsilon\, dX = \frac{1}{2}\int_{\Omega} \epsilon^\top \underbrace{\int_{\Omega^c} \bar{M}_2^\top C\, \bar{M}_2\, d\mathtt{X}^c}_{\mathcal{K}_\epsilon(\mathtt{X})}\, \epsilon\, d\mathtt{X}$$

Generalized linear elastic model

Replacing all the above expressions in Hamilton's principle produces:

$$\forall \mathtt{X} \in \Omega : \qquad \mathcal{M}\ddot{r} + \mathcal{F}^*(\mathcal{K}_\epsilon \mathcal{F} r) = 0 \qquad \text{(PDE)}$$

$$\forall \mathtt{S} \in \partial\Omega_N : \qquad F_\partial(\hat{n})\, \mathcal{K}_\epsilon \mathcal{F} r = \tau_N \qquad \text{(Neumann B.C.)}$$

$$\forall \mathtt{S} \in \partial\Omega_D : \qquad r = r_D \qquad \text{(Dirichlet B.C.)}$$

where:

- $r(\mathtt{X}, t)$: Generalized displacement vector.
- $\mathcal{M}(\mathtt{X})$: Mass density matrix.
- $\mathcal{K}_\epsilon(\mathtt{X})$: Strain-based stiffness density matrix.
- $\mathcal{F}$, $\mathcal{F}^*$: Linear differential operators.
- $F_\partial(\hat{n})$: Boundary-valued matrix operator induced by $\mathcal{F}$.

Other useful definitions:

- $\epsilon(\mathtt{X}, t) = \mathcal{F} r(\mathtt{X}, t)$ Generalized strain (kinematic equation).
- $e_\epsilon(\epsilon) = \mathcal{K}_\epsilon(\mathtt{X})\epsilon(\mathtt{X}, t)$ Generalized stress (constitutive equation).
- $\mathcal{K}_r(\cdot) = \mathcal{F}^*(\mathcal{K}_\epsilon(\mathtt{X})\mathcal{F}(\cdot))$ Displacement-based stiffness operator.

The end.